Highlights

Standard validation overestimates cross-basin landslide susceptibility transfer: a rigorously evaluated few-shot benchmark across 14 Chilean watersheds

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- Random cross-validation inflates cross-basin transfer F1 by 5-27 percentage points.
- Balanced test sets hide that precision at operational prevalence is only 0.07-0.10.
- Meta-learning beats Independent at K=1 in 4/4 targets, significant in 3/4.
- Episodic meta-training extracts a transferable geomorphic prior; naive pooling fails.
- Pooled logistic regression stays competitive: the finding is the prior itself.

Standard validation overestimates cross-basin landslide susceptibility transfer: a rigorously evaluated few-shot benchmark across 14 Chilean watersheds

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Abstract

Landslide susceptibility mapping in *newly-studied* basins is limited by the cost of local inventories, and the transfer of models from data-rich basins is routinely assessed with protocols that overstate it. We quantify that overstatement—random cross-validation inflates F_1 by 5–27 pp over spatially disjoint validation, balanced test sets hide that precision at operational prevalence falls to 0.07–0.10, and pooling correlated episodes narrows confidence intervals by 3.4×—and show that under rigorous evaluation a *transferable geomorphic prior* nonetheless exists, which episodic meta-training extracts where naive pooled pretraining fails. We benchmark first-order meta-learning (Reptile, FOMAML) against Independent, fine-tuning, domain-adversarial and classical baselines across 14 Chilean basins spanning ∼32° of latitude (∼1 to ∼1,900 mm/yr), evaluating K-shot transfer ($K \in \{1, 5, 10, 20\}$) on four targets from a 30 m MERIT-DEM stack. Under spatially disjoint validation, meta-learning outperforms Independent at $K = 1$ in all four targets (mean lift +5.5 to +6.2 percentage points F_1), significant under a cluster-level paired test in three of four; the smallest ($N = 19$) has too few spatial partitions to resolve it. The advantage decays monotonically and vanishes by $K = 20$, and absolute discrimination remains modest in two heterogeneous targets ($F_1 \approx 0.5$). A pooled-source logistic regression is competitive at every K: the contribution is the prior itself and the narrow few-shot regime where it pays off, not a new state-of-the-art predictor. Domain-adversarial adaptation improves on Independent in no target. The benchmark, datasets and protocol are released to support low-resource Earth-observation transfer in mountain-hazard monitoring.

Keywords: meta-learning, landslide susceptibility, transfer learning, remote sensing, spatial cross-validation, few-shot learning, Andes

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1. Introduction

Remote sensing has reshaped landslide susceptibility mapping by delivering globally consistent inputs: Sentinel-1 C-band SAR products for surface-deformation and amplitude features (Poggi et al., 2026), Sentinel-2 multi-spectral composites for surface cover and vegetation indices, MERIT and Copernicus 30 m DEMs for topographic and hydrologic derivatives, and WorldClim 2.1 bioclimatic surfaces for environmental context. These data, harmonized to a common grid, feed the dominant paradigm of fitting a machine-learning classifier to a per-basin inventory of labeled events—a workflow that produces strong results when local training data are abundant (Reichenbach et al., 2018; Merghadi et al., 2020; Ma et al., 2025), and that sits within the broader trajectory of deep-learning adoption across photogrammetric and remote-sensing tasks documented by Heipke and Rottensteiner (2020). However, the practitioner facing a *newly-studied* basin confronts a data-scarcity bottleneck: digitizing a usable inventory typically requires expert geomorphological mapping over hundreds of events, an effort that can take months and is rarely funded for individual watersheds. As a result, basins with sparse inventories ($N < 50$ mass-movement records) cannot benefit from data-hungry deep learning approaches, while transfer of models trained on data-rich neighboring basins suffers from systematic shifts in geology, climate, and surface cover (Zhu et al., 2026).

Two strategies have emerged to mitigate this. **Transfer learning** by fine-tuning a model pre-trained on data-rich basins (Ghorbanzadeh et al., 2021) reduces label requirements but still needs tens to hundreds of target samples. **Domain adaptation** approaches such as DANN (Ganin and Lempitsky, 2015) attempt to learn domain-invariant representations from labeled source data and unlabeled target data, with mixed success in remote sensing (Tuia et al., 2016; Huang et al., 2026); recent work on source-free domain adaptation for cross-region Sentinel time-series mapping further documents how p(x) and p(y|x) shifts dominate cross-site transfer (Mohammadi et al., 2024).

A complementary paradigm, **meta-learning** (Finn et al., 2017; Nichol et al., 2018), trains a model to *adapt rapidly to new tasks from a few examples*. Meta-learning has shown promise in remote sensing for land-cover classification (Rußwurm et al., 2020), satellite image segmentation (Tseng et al., 2023), and cross-scenario hyperspectral-to-Sentinel-2 trait estimation (Fu et al., 2025), and a recent single-site application (Ma et al., 2025) demonstrated temporal meta-learning for landslide susceptibility in Hong Kong. However, to our knowledge, no published study has systematically evaluated meta-learning for *cross-basin* landslide susceptibility transfer, particularly across the extreme hyper-arid-to-humid climate gradient of Chile, which spans $\sim 32^\circ$ of latitude and where traditional transfer learning is difficult. A second, more basic obstacle cuts across all of these strategies: how transfer is *measured*. Spatially explicit cross-validation, essential for unbiased evaluation on autocorrelated Earth-observation data (Brenning, 2005), remains under-applied in landslide remote sensing; benchmarks are routinely reported on balanced test sets at a fixed threshold, with episodes or folds pooled as though independent, and with negatives drawn far from the positives. Each of these conventions biases the reported skill upward, and their combined effect on *cross-basin* transfer has not, to our knowledge, been quantified. A transfer result that

survives only under permissive validation cannot support the operational claim it is usually made to carry.

In this work we quantify that gap and then ask what survives it. Using landslide susceptibility across a $\sim 32^\circ$ hydro-climatic gradient in Chile as the test bed, and first-order meta-learning as the transfer paradigm under scrutiny—the first systematic *cross-basin* evaluation of it, complementing the temporal single-site setting of Ma et al. (2025)—we measure how much each evaluation convention inflates apparent transfer, and re-examine the methods under the stricter protocol. Our contributions are:

1. We construct a unified, balanced dataset spanning **14 Chilean basins** (10 source, 4 target) with a consistent 58-feature cross-basin schema: 30 m MERIT-DEM derivatives (terrain, hydrology, texture, focal statistics), WorldClim bioclimatic variables, and SERNAGEOMIN geology categoricals. Inventory labels span four published catalogs and one ad-hoc digitization (3,290 positives + 3,290 negatives after balanced sampling).
2. We benchmark **five methods**—Independent (no source), Fine-tune (concat-source pretrain), Reptile, FOMAML, DANN—plus **four classical machine-learning baselines** (logistic regression, random forest, XGBoost, CatBoost), across K-shot evaluation ($K \in \{1, 5, 10, 20\}$) on four target basins, with three random seeds and bootstrap 95% confidence intervals (70,200 K-shot model fits in total across primary, classical and extended baselines).
3. We **quantify how much four standard evaluation conventions inflate apparent cross-basin transfer**, each isolated as a controlled contrast: random versus spatially disjoint validation (5–27 pp F_1), balanced test sets at a fixed threshold versus operational prevalence (precision falls to 0.07–0.10), pooling correlated episodes versus a cluster-aware bootstrap (confidence intervals 3.4× too narrow), and permissive versus slope-matched negative sampling. We further verify that a fixed adaptation budget does not favour the meta-learned models over the from-scratch baseline. Together these define an evaluation protocol for cross-basin transfer that we release with the benchmark.
4. Our findings: (i) meta-learning (Reptile or FOMAML) wins $K = 1$ over Independent in 4/4 target basins under spatial CV, with a mean FOMAML lift of $+5.5$ pp F_1 (range $+4.3$ to $+6.8$ pp; Reptile mean $+6.2$ pp, up to $+7.8$ pp), statistically significant under a cluster-level paired test in three of the four targets; Reptile and FOMAML perform within overlapping confidence intervals in the majority of target-K cells; (ii) the advantage decays monotonically with K and vanishes by $K = 20$; (iii) DANN does not improve over Independent on average and degrades substantially in the smallest target basin (Copiapó $N = 19$: -7 pp at $K = 1$), suggesting adversarial alignment is counter-productive when both source domains and target inventories are small; (iv) adaptation curves confirm meta-learned initializations reach $F_1 \approx 0.78$ within ≤ 3 gradient steps in Copiapó versus > 100 steps for Independent; (v) a pooled-source logistic regression is competitive with meta-learning in peak F_1 at every K tested, so the benchmark's central claim is not predictive superiority: it is the mechanistic evidence that a transferable geomorphic prior exists—episodic meta-training extracts it

within a neural architecture where naive pooled pretraining (Fine-tune, Meta-Baseline) fails—plus a quantification of the narrow few-shot regime where that prior pays off.

The remainder of this paper is organized as follows. Section 2 reviews related work in landslide ML, transfer learning, and meta-learning. Section 3 describes the study area and data. Section 4 details the methodology. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Landslide susceptibility modeling

Statistical and machine-learning approaches to landslide susceptibility have been the subject of extensive review (Reichenbach et al., 2018; Merghadi et al., 2020). The field has progressively moved from bivariate and weights-of-evidence approaches to logistic regression, random forests, gradient-boosted trees (XGBoost, CatBoost), and more recently to deep neural networks. Brenning (2005) established spatial cross-validation as a methodological best practice for landslide modeling, demonstrating that random hold-out validation systematically overestimates generalization error due to spatial autocorrelation between nearby training and test pixels. Object-oriented approaches (Stumpf and Kerle, 2011) and patch-based convolutional models have refined the spatial feature representation, but the fundamental requirement of hundreds to thousands of labeled events per study area remains a binding constraint. Recent benchmarks comparing classical and ensemble ML algorithms (Merghadi et al., 2020) report F_1 scores in the 0.75–0.90 range when $N > 200$, but performance degrades sharply for small inventories. Most of the field operates within a single basin or contiguous region; cross-basin transfer is studied less frequently and is typically framed as out-of-sample testing rather than as a problem requiring methodological adaptation.

2.2. Transfer learning in remote sensing

Transfer learning by fine-tuning a model pre-trained on data-rich source domains is the de facto strategy for label-efficient remote sensing (Tuia et al., 2016). In landslide applications, Ghorbanzadeh et al. (2021) demonstrated that ImageNet-pretrained convolutional networks fine-tuned on a few hundred local samples outperform train-from-scratch baselines, confirming that representations learned on natural images contain transferable low-level features. However, fine-tuning a deep network with $K < 50$ target labels typically results in catastrophic over-fitting unless aggressive regularization or early stopping is applied. Furthermore, the "concat-source" variant—pre-training on a union of source basins, which is the baseline we adopt—implicitly assumes that all source domains are equally relevant to the target, a hypothesis that is rarely tested explicitly and that fails when the source pool spans heterogeneous environmental conditions. Recent work has begun to formalize cross-region transferability through spatiotemporal deep-learning frameworks that integrate environmental context (Luo et al., 2026), transferable landslide segmentation via universal adapters (Wei et al., 2024), cross-resolution landslide mapping (Ng et al., 2025), and adversarial domain

adaptation (Zhang et al., 2025); semi-supervised event detection from multi-temporal optical imagery offers a complementary path for sparse-inventory settings (Deijns et al., 2024). However, these approaches target dense segmentation from imagery, and landslide-specific cross-basin meta-learning on point inventories with rigorous spatial validation remains an open question.

2.3. Domain adaptation

Domain adaptation (DA) seeks to learn classifiers that generalize across distribution shifts by leveraging both labeled source data and unlabeled target data. The Domain-Adversarial Neural Network (DANN) of Ganin and Lempitsky (2015) is the canonical method: a feature extractor is trained simultaneously to (i) discriminate the class label and (ii) confuse a domain classifier through a Gradient Reversal Layer, encouraging the learned representation to be domain-invariant. Variants based on Maximum Mean Discrepancy (Long et al., 2015) and Wasserstein distance have followed. In remote sensing, DA has been applied to multi-source land-cover classification, image segmentation, and scene recognition, with mixed success (Tuia et al., 2016): performance gains are typically substantial when source domains are abundant and labeled target data are completely absent, but degrade when only a handful of source domains are available or when small numbers of labeled target samples are accessible. The latter regime is precisely our setting, and our results below confirm that DANN is not a competitive baseline when source-domain count is small.

2.4. Meta-learning

Meta-learning frames the data-scarcity problem as one of *learning to learn*: the model is trained on a distribution of related tasks so that adaptation to a new task from few examples requires only a small parameter update. Model-Agnostic Meta-Learning (MAML; Finn et al. 2017) is the canonical second-order method, computing gradients through inner adaptation steps, but it is computationally expensive and unstable in practice. First-order approximations—FOMAML, which drops the second-order term, and Reptile (Nichol et al., 2018), which replaces query-loss-driven updates with averaged inner-step displacements—are more practical. Theoretical work (Raghu et al., 2020) suggests that MAML's success can be largely attributed to feature reuse rather than rapid weight adaptation, which we revisit in our adaptation-curve analysis (Section 5).

In remote sensing, meta-learning has been applied to few-shot land-cover classification (Rußwurm et al., 2020), achieving 70–90% accuracy with $K = 5$ samples per class on EuroSAT and Sen12MS benchmarks, and to satellite image classification more broadly (Tseng et al., 2023). Ma et al. (2025) recently demonstrated temporal meta-learning for dynamic landslide susceptibility mapping at a single site (Hong Kong), decomposing yearly subtasks and achieving fast adaptation in 5 samples and 5 inner steps. Our work complements this in two ways: (i) we focus on *spatial* cross-basin transferability rather than temporal adaptation; (ii) we evaluate on 14 basins across a 1500 km climate gradient, providing the first cross-basin benchmark of first-order meta-learning algorithms (Reptile, FOMAML) for landslide susceptibility under rigorous spatial cross-validation.

3. Study area and data

3.1. Basins

We study 14 watersheds distributed along the length of Chile (Figure 1), from the hyper-arid Atacama Desert (∼21°S) to Patagonia (∼53°S), selected to span an extreme hydro-climatic gradient (basin-mean annual precipitation ∼1 to ∼1,900 mm/yr, with individual pixels in the wettest basin exceeding 2,500 mm/yr) and to have heterogeneous landslide-mapping coverage. **Ten source basins** are used for meta-training: Chañaral / Río Salado (BNA 032, hyper-arid; 1,710 events from Parra (2025) supplementary data), Costeras Q. Negra–Pan de Azúcar / Taltal (BNA 029, hyper-arid coastal; 80 mass-movement polygons + 300 negatives from Parra (2025)), Costeras Tilviche-Loa (BNA 018; 127 events), Costeras Loa-Caracoles (BNA 022; 50 events), Río Maipo (BNA 057; 349 events), Río Rapel (BNA 060; 392 events), Río Maule (BNA 073, semi-humid; the 203 of the 1,227 events in the 2010 M_w 8.8 earthquake-induced inventory of Serey et al. (2019) that fall within the basin polygon, retaining all four mapped typologies—disrupted slides, flows, lateral spreads and coherent slides), Río Choapa (BNA 047, semi-arid; 62 events from the SERNAGEOMIN national *catastro de remociones en masa*), Costeras Bueno-Puelo (BNA 104; 47 events), and Costeras Magallanes (BNA 125; 413 events). **Four target basins** are used for K-shot evaluation: Río Copiapó (BNA 034; 19 events), Río Huasco (BNA 038; 278 events), Río Elqui (BNA 043; 132 events), and Río Limarí (BNA 045; 35 events). The split between source and target is motivated both by data availability and by ensuring climate-gradient coverage in both groups.

3.2. Remote-sensing and geospatial data sources

All conditioning factors are derived from open Earth-observation products harmonized to a common 30 m raster grid in UTM zone 19S. Table 1 summarizes the five primary sources, their spatial and temporal coverage, and the feature families derived from each.

The satellite-derived MERIT DEM and its terrain, hydrology and texture derivatives supply the bulk of the stack. MERIT DEM is preferred over individual SRTM tiles because its hydrologic corrections (pit filling, stream burning) yield more stable flow-routing derivatives essential for HAND and sediment connectivity. WorldClim 2.1 captures the 1970–2000 climatological baseline; while static climate features cannot represent extreme events that trigger individual landslides, they parameterize the long-term hydro-climatic regime that conditions susceptibility (Ma et al., 2025). All raster inputs are resampled bilinearly (continuous variables) or by nearest neighbor (categorical), then stacked into a per-pixel feature vector consumed by the classifier. We acknowledge that this introduces an effective-resolution mismatch: MERIT-derived terrain and texture features inherit the ∼90 m source resolution despite being sampled on the 30 m grid, and WorldClim bioclimatics are nearly piecewise-constant within each ∼5 km native cell. Topographic and texture derivatives can thus be expected to be smoother than a native 30 m DEM (e.g., Copernicus GLO-30) would yield, and the climate-distance analysis of Section 5.8 is interpreted at basin level rather than per pixel. We return to this limitation in Section 6.6. Intermediate per-basin products can be regenerated from public sources using the released pipeline (see the Data and code availability statement at the end of the manuscript) to support reproducibility.

Table 1: Remote-sensing and geospatial inputs used to compute the per-basin feature stack. All products are public and openly accessible. Sentinel-2 optical composites were computed for two source basins only and are excluded from the common cross-basin schema (Section 3.3); the operative inputs are the satellite-derived MERIT DEM, its terrain/hydrology/texture derivatives, WorldClim climate and SERNAGEOMIN geology.

Product	**Sensor / Source**	**Native res.**	**Period**	**Features derived**
MERIT DEM (filled, hydrology-corrected)	SRTM + AW3D + ICESat fusion	3 arc-sec (∼90 m)	static	28 terrain features (slope, aspect, MRVBF, openness, geomorphons, etc.); 10 hydrology features (flow accumulation, HAND, TWI, drainage density, sediment connectivity); 6 GLCM texture; 6 focal statistics
WorldClim 2.1	Multi-station interpolated climatology	2.5 arc-min (∼5 km)	1970–2000 baseline	5 bioclimatic variables (bio_01 mean temp; bio_12 annual precip; bio_13/14 wettest/driest month precip; bio_15 seasonality)
SERNAGEOMIN national geological map	National Geological Survey of Chile	1:1,000,000 vector	static	3 categorical features (lithology, rock type, geological age)
BNA basin polygons + landslide inventories	DGA + SERNAGEOMIN *catastro de remociones* + Serey et al. (2019) + Parra (2025)	vector	1922–2023 mixed	Basin extent for sampling; positive labels (3,290) and negatives (3,290)

3.3. Conditioning factors

We compute a common cross-basin stack of **58 features** per basin at 30 m grid alignment, organized in six categories. To guarantee an identical feature schema across all 14 basins, we retain only features available in every basin (Section 3.4); Sentinel-2 spectral composites, computed for two coastal source basins (Taltal, Choapa) but not the remaining twelve, are therefore excluded by this intersection and do not enter the benchmark. The 58 retained features are:

- **Terrain (28):** slope, aspect, eastness, northness, hillshade, sky-view factor, openness (positive, negative), geomorphons, landform, Multi-resolution Index of Valley Bottom Flatness (MRVBF), Multi-resolution Index of Ridge-Top Flatness (MRRTF), vector ruggedness measure, deviation from mean elevation, convergence, curvature, curvedness, shape index, surface area ratio, three TPI radii plus TPI, valley depth, relative slope position, LS-factor, TRI.

- **Hydrology (10):** filled DEM, flow accumulation (D8, MFD), flow direction (D8, D∞), HAND, stream network, TWI, drainage density, sediment connectivity (Borselli et al., 2008).
- **Texture GLCM (6):** contrast, correlation, dissimilarity, energy, entropy, homogeneity, computed on the DEM rather than on optical imagery so that the texture descriptors capture topographic roughness (a physical correlate of slope instability) and remain available in every basin, independent of cloud-free optical coverage.
- **Focal statistics (6):** DEM std at radii 1, 4, 10; DEM range $r = 4$; slope std $r = 4$; slope mean $r = 4$.
- **Climate (5):** WorldClim 2.1 bioclimatics—bio_01 (annual mean temperature), bio_12 (annual precipitation), bio_13/14 (precipitation wettest/driest month), bio_15 (precipitation seasonality).
- **Geology (3):** SERNAGEOMIN 1:1,000,000 lithology class, rock type, geological age (categorical, encoded).

All factors are computed using SurtGIS, a Rust-based geospatial toolkit (Parra, 2026), with reprojection to UTM zone 19S and bilinear/categorical resampling as appropriate.

3.4. Sampling protocol

We adopt a **hybrid balanced sampling** strategy. Each pixel of the 30 m DEM grid is the unit of analysis. For each basin:

1. **Positives:** For point inventories, snap to nearest pixel. For polygon inventories (Taltal), rasterize with `all_touched=True` and randomly sub-sample at most 30 pixels per polygon to limit spatial autocorrelation.
2. **Negatives:** When the inventory provides explicit non-mass-movement points (Chañaral, Taltal), we use them and expand each by a 60 m buffer to cover surrounding pixels. When negatives are unavailable, we generate them by random sampling within the basin polygon, excluding a 500 m buffer around positives to avoid spatial leakage.
3. **Balancing:** Negatives are sub-sampled or augmented to match positive count exactly (1:1).

This procedure yields 9,182 samples across the 14 basins: 3,290 positives and 3,290 negatives form the balanced cross-validation pool, complemented by 2,602 additional source-only negatives used exclusively during meta-training (target basins contribute no additional source-only samples to avoid leakage into K-shot evaluation).

3.5. Feature selection via correlation-based feature selection (CFS)

We apply **correlation-based feature selection** (Hall, 1999) using symmetric uncertainty (information-theoretic, robust to mixed data types). Features are quantile-discretized into 10 bins. Forward greedy search adds the feature that maximizes the subset score:

$$M_S = \frac{k \cdot \overline{r_{cf}}}{\sqrt{k + k(k-1) \cdot \overline{r_{ff}}}}, \tag{1}$$

where k is subset size, $\overline{r_{cf}}$ the mean feature-class correlation, and $\overline{r_{ff}}$ the mean inter-feature correlation. The procedure terminates when no feature addition improves the score.

CFS selects **11 features**: 6 terrain (landform, openness positive, MRVBF, MRRTF, curvedness, geomorphons), 2 focal statistics (DEM std $r = 10$, slope std $r = 4$), 2 texture GLCM (entropy, correlation), and 1 climate (bio_12 annual precipitation). Notably, no hydrology or geology features are selected, and annual precipitation (bio_12) is the sole non-DEM feature retained—terrain ruggedness and texture dominate the discriminative signal across the Chilean landslide inventories.

4. Methods

4.1. Backbone model

All methods share a 3-layer MLP backbone: 11 input features → 64 hidden → 32 hidden → 1 output, with ReLU activations, 10% dropout per hidden layer, and binary cross-entropy with logits as the training loss. We deliberately use a small tabular network rather than a CNN or Transformer because (i) our feature representation is point-wise and scale-invariant, (ii) our pixel-level samples lack spatial neighborhoods at this stage of the pipeline, and (iii) it ensures interpretability of feature importance. Section 6 discusses the natural extension to U-Net architectures for spatially-structured prediction.

4.2. Methods compared

Independent. Train MLP from scratch on K target labels (K positives + K negatives), 30 Adam steps at learning rate 10^{-2}. No source-basin information is used.

Fine-tune (concat-source baseline). Pretrain MLP on the concatenation of all 10 source basins (300 epochs at 10^{-3}), then fine-tune for 30 steps on K target samples.

Reptile. (Nichol et al., 2018). First-order meta-learning. For 300 outer steps:

1. Sample a source basin τ.
2. Sample $K_{\text{meta}} = 10$ support points.
3. Adapt model with 5 inner SGD steps at 10^{-2}.
4. Update meta-model: $\theta \leftarrow \theta + \epsilon(\theta_{\text{adapted}} - \theta)$ with $\epsilon = 0.1$.

At test time: load meta-model and adapt for 30 steps on K target samples (same protocol as Fine-tune).

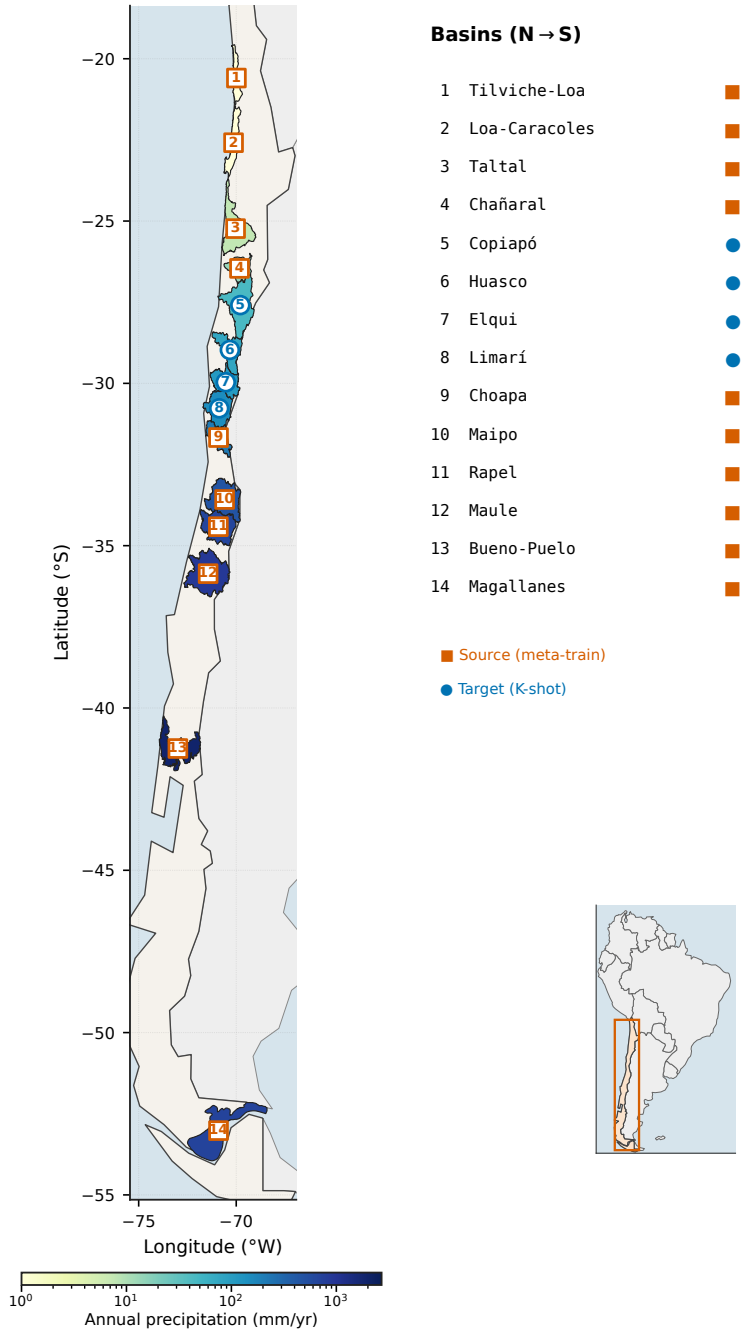


Figure 1: Fourteen Chilean watersheds spanning an extreme climate gradient—from the hyper-arid Atacama in the north (∼1 mm/yr) to the humid Patagonian Andes in the south (∼1,900 mm/yr basin mean), over ∼32° of latitude. Basins are numbered north-to-south; source basins (orange squares) are used for meta-training and target basins (blue circles) for K-shot evaluation. Polygons are color-coded by mean annual precipitation (WorldClim bio_12, log scale). Inset: location within South America.

FOMAML (First-Order MAML). For 300 outer steps:

1. Sample task τ, support S, query Q.
2. Inner: 5 SGD steps on S, lr 10^{-2}.
3. Compute query loss on the *adapted* model.
4. Take query-loss gradient w.r.t. adapted parameters; copy this gradient to the meta-model.
5. Apply Adam outer-step at lr 10^{-3}.

DANN. (Ganin and Lempitsky, 2015). Feature extractor F: $11 \to 64 \to 32$ (shared). Class head C: $32 \to 1$. Domain head D: $32 \to 10$ (with Gradient Reversal Layer). Trained for 100 epochs at batch 64, lr 10^{-3}, with annealed $\lambda(p) = 2/(1 + e^{-10p}) - 1$. After training, $F + C$ are loaded into the MLP backbone for K-shot adaptation following the same protocol as Fine-tune.

ProtoNet. (Snell et al., 2017). Prototypical networks. A shared feature encoder (same $11 \to 64 \to 32$ architecture as the MLP backbone) is meta-trained episodically: for each source task, class prototypes are the mean support embeddings, and the prototypical loss is the cross-entropy of negative squared-Euclidean distances from query embeddings to prototypes (300 outer steps, Adam at 10^{-3}). At test time, the K target support points define prototypes and query pixels are classified by nearest prototype—no gradient adaptation is required.

Meta-Baseline. (Chen et al., 2021). The encoder is first pre-trained with standard supervised classification on the concatenated source pool, then frozen; at test time, K-shot cosine-similarity prototypes classify query pixels. This tests the claim that a strong pre-trained representation with a simple metric head can match episodic meta-learning.

CDAN. (Long et al., 2018). Conditional adversarial domain adaptation. The domain classifier is conditioned on the outer product of feature embeddings and class-probability predictions (rather than features alone as in DANN), which the authors argue yields better-aligned multimodal distributions. Training and K-shot adaptation follow the same protocol as DANN.

Spatial CNN (U-Net-style) backbone. As a methodological control on the point-wise MLP, we also train a small patch-classifier CNN on multi-channel image patches. Standard dense-segmentation benchmarks are not directly applicable to sparse point-inventory susceptibility, which has neither pixel-level masks nor bounding boxes; we therefore adopt a patch-classifier formulation that predicts per-point susceptibility from a local feature window. For each inventory point we extract a 32×32 window (covering $\sim 960 \times 960$ m at 30 m resolution) from the 25 gridded feature rasters (terrain + hydrology + texture + focal statistics) consistently available across all basins. The architecture is three Conv–ReLU blocks (32, 32, 64 channels) with two MaxPool layers, an adaptive global average pool, and a 32-unit FC head producing a binary logit ($\sim$ 74,000 parameters). Reptile, FOMAML, Fine-tune (concat-source pre-train) and Independent are applied to this backbone under the same spatial K-shot CV protocol on the largest target basin (Huasco), with five fewer episodes per fold (15 vs. 20) to

keep compute bounded. Magallanes was excluded from the source pool for this experiment because its raster stack exceeds the worker memory budget. Results (Section 5.12) quantify the value of spatial context for sparse-inventory susceptibility transfer.

Classical ML baselines. For comparison, we evaluate four classical algorithms under identical K-shot spatial-CV protocol: **logistic regression** (sklearn, L_2 regularization, $C = 1.0$, 500 iterations); **random forest** (100 estimators, max depth 10); **XGBoost** (100 estimators, max depth 5, lr 0.1); **CatBoost** (100 iterations, depth 5, lr 0.1). Each algorithm is run in two modes: *independent* (trained only on K target samples) and *concat-source* (trained on full source pool + K target samples).

Statistical protocol for extended baselines. ProtoNet, Meta-Baseline, and CDAN are evaluated under the same spatial-CV protocol as the primary methods but with **five seeds** $\{42, 123, 7, 1729, 2718\}$ (vs. three for the main benchmark) to strengthen statistical power, yielding 16,380 additional K-shot adaptations.

4.3. K-shot evaluation protocol

Throughout, K denotes the number of support examples *per class*: each K-shot episode uses K labelled positives and K labelled negatives ($2K$ support samples in total), so that $K = 1$ corresponds to a single verified mass-movement footprint plus one non-landslide pixel. For each (target basin, K, method, seed) combination, we run two evaluation protocols:

Random cross-validation. K-shot episodes with support and query sampled at random from the basin's pixel pool; spatial proximity is not enforced. The realized benchmark comprises 40 episodes per (target, K, method, seed); Copiapó is restricted to $K \in \{1, 5\}$ by its 19-event inventory, yielding 8,400 adaptations in total.

Spatial cross-validation. The target basin is partitioned into 5 spatial clusters via K-means on (x, y) UTM coordinates. For each fold (held-out cluster), 20 episodes sample K positives + K negatives from the *training* clusters as support; the query is the *full held-out cluster*. This guarantees support and query are spatially disjoint, addressing spatial autocorrelation per Brenning (2005); Pohjankukka et al. (2017); Valavi et al. (2019). Fold × seed partitions whose held-out cluster lacks both classes, or whose training clusters cannot supply K positives and K negatives, yield no valid episodes and are skipped; the number of valid combinations is therefore basin-dependent (Elqui 15/15, Limarí 12/15, Copiapó 9/15, Huasco 6/15). We acknowledge that K-means folds do not enforce a block size larger than the empirical variogram range of model residuals; per the framework of Meyer and Pebesma (2021), this implies that reported metrics describe interpolation within the basin's training extent rather than extrapolation. A formal area-of-applicability analysis is left as an avenue for follow-up work.

Each episode:

1. Sample support S (K positives + K negatives).
2. For meta-trained methods (Reptile, FOMAML, DANN), load the corresponding meta-model; otherwise initialize from scratch.

3. Adapt with 30 Adam steps on S.
4. Evaluate F_1 and ROC-AUC on the query.

Reported F_1 / AUC are bootstrap means with 95% confidence intervals across 1,000 resamples of all (seed × episode) results pooled. F_1 is computed at a fixed decision threshold of 0.5 on *balanced* (1:1) queries; because operational susceptibility mapping faces extreme class imbalance, this balanced F_1 overstates the precision a practitioner would observe. We therefore also report the threshold- and prevalence-independent ROC-AUC in the main text, and a dedicated prevalence-robust analysis (precision and average precision re-weighted to realistic landslide prevalences; Section 5.3).

Cluster-robust uncertainty. Pooling episodes as independent understates uncertainty: within one (seed, fold) spatial partition, all 20 episodes evaluate on the *same* held-out query cluster and draw supports from the same small training pool, so the effective sample size is the number of (seed, fold) clusters (6–15 per target), not the nominal 120–300 runs. We therefore report two additional cluster-aware quantities for the headline $K = 1$ comparison. First, a **two-stage hierarchical bootstrap** resamples clusters with replacement, then episodes within each resampled cluster (2,000 resamples), widening the per-method CI to reflect the true replication level. Second, a **cluster-level paired test**: because every method is evaluated on the same partitions, the per-cluster mean-F_1 difference between a method and Independent is a valid paired observation; we report its mean, its hierarchical-bootstrap 95% CI, and a two-sided Wilcoxon signed-rank test on the cluster-level differences (Table 4). This paired test exploits the shared spatial partitions and is the appropriate significance criterion; the marginal per-method CIs (which ignore the pairing) are shown only for completeness.

The main spatial-CV K-shot benchmark (Table 2) nominally spans 4 targets × 4 K values × 5 methods × 3 seeds × 5 folds × 20 episodes = 24,000 adaptations; after excluding Copiapó $K = 20$ and infeasible fold × seed partitions, the realized benchmark comprises 15,900 K-shot adaptations. Adding the random-CV benchmark (8,400 adaptations), the classical-ML baselines (29,520 fits under both CV protocols and both train modes), and the extended meta-learning baselines (16,380 adaptations; Section 4.2) yields 70,200 K-shot model fits, complemented by the adaptation-curve experiment and the hyperparameter-sensitivity ablation grid.

5. Results

5.1. Random-CV K-shot benchmark

Under random cross-validation (Figure 2), meta-learning wins $K = 1$ in all four target basins, with best-meta-learner advantages of +5.4 to +8.8 percentage points (pp) F_1 over Independent: FOMAML leads in Huasco, Reptile leads in Elqui and Limarí, and the two are tied in Copiapó ($F_1 = 0.675$ both). The advantage narrows to 0–2 pp at $K = 10$ and disappears at $K = 20$, where all methods converge to within 1 pp F_1. This pattern is consistent with the theoretical expectation that meta-learning's advantage scales inversely with available task-specific data.

5.2. Spatial-CV K-shot benchmark

Under spatial cross-validation—where support and query are sampled from disjoint K-means clusters of the basin—**meta-learning (Reptile or FOMAML) wins $K = 1$ in 4/4 target basins** (Figure 3, Table 2). Reptile achieves the nominally highest F_1 at $K = 1$ in all four targets (Copiapó 0.665, Huasco 0.755, Elqui 0.476, Limarí 0.528), with FOMAML statistically tied in each (Copiapó 0.663); the inter-method gap between Reptile and FOMAML is within bootstrap standard error in every case. FOMAML achieves +5.5 pp F_1 mean advantage over Independent across targets at $K = 1$ (range +4.3 to +6.8 pp; Reptile mean +6.2 pp, up to +7.8 pp in Huasco), decaying monotonically to −1.1 pp at $K = 20$ (over the three basins where $K = 20$ is feasible); Reptile retains +1.9 pp at $K = 20$, within CI overlap of Independent. Fine-tune provides essentially no benefit (within ±1 pp).

We assess the significance of the $K = 1$ advantage with the cluster-level paired test described in Section 4.3 (Table 4), which respects the fact that episodes within a spatial partition are not independent. The 95% hierarchical-bootstrap CI of the per-cluster paired difference (meta-learning − Independent) **excludes zero in three of the four targets**: Huasco (Reptile +7.8 pp [+4.8, +11.6], all 6/6 clusters positive, Wilcoxon $p = 0.03$), Elqui (+4.7 pp [+2.6, +6.9], 13/15, $p = 0.001$) and Limarí (+6.7 pp [+3.4, +9.8], 10/12, $p = 0.007$). In Copiapó the advantage is of similar magnitude but *not* statistically resolved (+5.7 pp [−0.1, +12.9], 7/9 clusters positive, $p = 0.16$): its 19-event inventory yields only nine (seed, fold) clusters, too few to separate a ~ 6 pp effect from noise. Neither Fine-tune nor DANN is significant in any target. The naive flat bootstrap that pools episodes as independent understates CI width by a median factor of 3.4× (6.4 vs. 20.7 pp), and its marginal per-method CIs should not be read as a significance test.

The spatial CV protocol yields lower absolute F_1 magnitudes than random CV (drop of 1 pp in Huasco to 27 pp in Elqui; Figure 5), confirming that random CV overestimates absolute performance due to spatial autocorrelation, but *the relative ranking of methods is preserved*.

5.3. Prevalence-robust evaluation

Because the balanced F_1 of the main benchmark is computed on 1:1 queries at threshold 0.5, it does not reflect the extreme class imbalance of operational mapping. We therefore assess the $K = 1$ advantage with two prevalence-aware lenses (Figure A.9). First, the **ROC-AUC**, which is threshold- and prevalence-independent: under spatial CV at $K = 1$, Reptile exceeds Independent in all four targets—Copiapó 0.805 vs. 0.638, Huasco 0.780 vs. 0.668, Elqui 0.745 vs. 0.649, Limarí 0.559 vs. 0.527 (Table 2 reports the corresponding F_1)—so the discrimination gain is not an artifact of the balanced operating point. Second, we translate the balanced operating characteristics to realistic prevalence via $\text{precision}(\pi) = \pi\,\text{TPR}/[\pi\,\text{TPR} + (1-\pi)\,\text{FPR}]$ and recompute the precision–recall curve at $\pi = 0.05$ (a plausible fraction of hazard-prone terrain; the raw pixel-level inventory prevalence is far lower). At this prevalence *absolute precision collapses for every method*—at $\pi = 0.05$ the precision at threshold 0.5 is only 0.087 (Reptile) vs. 0.069 (Independent) in Huasco, i.e. most flagged pixels are false positives—a direct, honest consequence of rare-event detection that the balanced F_1 masks. Crucially, however, **the meta-learning advantage survives the**

re-weighting and is proportionally larger: the average precision at $\pi = 0.05$ favours Reptile over Independent in all four targets (Copiapó 0.130 vs. 0.066, a near-doubling; Huasco 0.108 vs. 0.087; Elqui 0.102 vs. 0.084; Limarí 0.060 vs. 0.048). The operational reading is twofold: the framework is a terrain-*stratification* tool (its value is in ranking, captured by AUC/AP, not in per-pixel classification), and the meta-learned prior's benefit is robust to prevalence, not an artifact of balanced evaluation.

5.4. Susceptibility maps: what the $K = 1$ lift looks like on the ground

To translate the benchmark advantage into the product a basin manager would actually use, we apply the $K = 1$-adapted models to the *full* 30 m grid of two targets—Copiapó (largest AUC/AP gain, Section 5.3; $N = 19$) and Elqui (a hard, low-signal basin)—and map the mean adapted-model output across the three seeds and twenty support draws (Figure 6). Maps are shown as within-basin percentiles, the standard susceptibility-class presentation, because the raw $K = 1$ outputs are uncalibrated (Section 6.6). In Copiapó, both models produce spatially coherent surfaces that track the drainage network and steep valley flanks, and the 19 inventory events fall predominantly in the upper susceptibility percentiles; the meta-learned (Reptile) surface sharpens the ridge–valley contrast and pulls the western headwater tributaries—where several inventory points cluster—into higher percentiles than the Independent baseline, consistent with its +5.7 pp F_1 edge. In Elqui, the maps make the low $F_1 \approx 0.48$ tangible: susceptibility is diffuse, the strong west–east orographic gradient dominates the pattern, and inventory events are scattered across percentile classes rather than concentrated in a coherent high-susceptibility zone. The map view thus corroborates both the headline result (the meta-learned prior yields a usable first-pass susceptibility layer from a single seed event) and its honest limit (in heterogeneous basins the layer is a screening product, not a planning-grade map).

5.5. Domain adaptation: DANN and CDAN both under-perform

DANN does not improve over Independent on average and shows a strongly basin-dependent pattern (Table 2). The under-performance is concentrated in Copiapó, the smallest-inventory target ($N = 19$ events), where DANN trails Independent by 5–7 pp F_1 across all K values. In the other three targets, DANN is roughly neutral or marginally positive: Huasco ±1 pp, Elqui +0.8 to +1.6 pp, Limarí −0.2 to +3.2 pp; at $K = 20$, DANN even wins Limarí (0.598, best of row).

To verify this is not an artifact of DANN's decade-old design, we additionally evaluate **CDAN** (Long et al., 2018), a stronger conditional-adversarial method (Table 3). CDAN exhibits the same failure mode: in Copiapó it reaches only $F_1 = 0.573$ at $K = 1$ (−3.6 pp vs. Independent's 0.609), and across all targets it remains below the meta-learning methods at $K = 1$ (Copiapó 0.573, Huasco 0.707, Elqui 0.440, Limarí 0.433). That a modern conditional-adversarial method reproduces the negative-transfer pattern indicates the problem is intrinsic to adversarial domain alignment under scarce source domains, not specific to the original DANN formulation. We hypothesize that adversarial alignment requires both a moderately-sized source pool *and* sufficient target signal to break ties between candidate

invariances; the conjunction of few source domains ($N = 10$) and a very small target inventory makes the gradient-reversal mechanism susceptible to negative transfer. We return to this in the Discussion (Section 6).

5.6. Adaptation curves confirm rapid convergence

FOMAML's meta-learned initialization achieves $F_1 \approx 0.78$ within ≤ 3 gradient steps in Copiapó (Figure 4), confirming that the prior is already discriminative and the few-shot inner adaptation is rapid. Reptile reaches its peak in 3–5 steps, Fine-tune in ~ 10 steps. Independent, trained from random initialization in the adaptation-curve experiment, requires > 100 steps before reaching a comparable plateau (well above the 30-step inner budget used in the main benchmark, Section 4.3). This validates the practical efficiency benefit of meta-learning: a pre-trained meta-model is immediately deployable for new basins, whereas Independent training requires complete model retraining from random initialization.

Because the main benchmark fixes the adaptation budget at 30 inner steps for *every* method, the slower convergence of Independent raises a fairness question: does the fixed budget under-serve the baseline and thereby inflate the reported lift? A budget-fairness control answers no. Re-running the $K = 1$ spatial-CV protocol with inner-step budgets of 30, 100 and 300 for both Independent and Reptile (shared support draws per episode), Independent gains at most +0.4 pp F_1 from a 10× larger budget (Copiapó 0.598 → 0.601, Huasco 0.729 → 0.730, Elqui 0.441 → 0.441, Limarí 0.452 → 0.456): with a two-sample support, additional steps saturate or over-fit rather than converge further on held-out queries—the > 100-step plateau above concerns the *trajectory* of training-loss descent, not the attainable query F_1. Measured against Independent at its own best budget, the Reptile lift remains +2.0 to +6.7 pp across the four targets (`results/stats/baseline_budget.csv`), so the fixed 30-step budget does not manufacture the meta-learning advantage.

5.7. Hyperparameter sensitivity

Sensitivity to inner steps $\{1, 3, 5, 10\}$, Reptile $\epsilon \in \{0.05, 0.1, 0.3\}$, and $K_{\text{meta}} \in \{5, 10, 20\}$ is small (within 0.5 pp; Figure A.12). The default configuration (inner steps = 5, $\epsilon = 0.1$, $K_{\text{meta}} = 10$) is robust.

5.8. No detectable relation between precipitation similarity and transfer success

We tested whether source-target climate similarity (measured as $|\text{bio_12}_{\text{source}} - \text{bio_12}_{\text{target}}|$) predicts FOMAML transfer advantage at $K = 5$ (Figure A.15). The correlation is negligible (Pearson $r = 0.01$, $p = 0.99$; Spearman $\rho = 0.20$, $p = 0.80$). Two qualifications keep this result in its place: with only four targets the test is severely underpowered, and $|\Delta\text{bio_12}|$ is a single-variable proxy (annual precipitation), not a full climatic or lithological distance; the weak positive Spearman trend cannot be excluded. What the data do support is the narrower statement that *transfer success is not dominated by precipitation similarity at this scale*—a preliminary indication, not established climate invariance, that climate-matched source selection is not the first-order design concern.

5.9. Classical ML is competitive under spatial CV

Logistic regression, random forest, XGBoost and CatBoost achieve competitive F_1 under spatial CV (Figure A.14). Logistic regression wins 7/16 (target, K) combinations; random forest wins 4/16. Meta-learning (FOMAML or Reptile) wins 5/16, principally at $K = 1$ in Copiapó and Limarí. We state the strongest classical configuration plainly: logistic regression trained on the pooled source basins plus the K target samples (concat-source mode) is competitive with meta-learning *already at* $K = 1$—Copiapó 0.696 vs. Reptile 0.665, Huasco 0.774 vs. 0.755, Elqui 0.473 vs. 0.476, Limarí 0.516 vs. 0.528—so on peak F_1 alone a practitioner has no reason to prefer meta-learning over a pooled logistic regression at any K we tested. What the neural-network family comparison isolates, however, is the *mechanism*: within the same architecture, episodic meta-training transfers where naive pooling does not (Fine-tune gains ±1 pp over Independent; the frozen concat-pre-trained Meta-Baseline fails outright, Section 5.10), which is precisely the evidence that a transferable geomorphic prior exists and that *how* it is extracted matters. The operational reading is honest: pooled logistic regression is a legitimate, simpler default for basin-scale susceptibility screening; the meta-learned initialization adds value in the deployment regime (an initialization that specialises in ≤ 3 gradient steps, Section 5.6) and as the instrument that demonstrates cross-basin transferability of the prior.

5.10. Modern meta-learning baselines confirm the advantage is method-robust

To test whether the few-shot advantage is specific to first-order methods (Reptile, FOMAML) or general to meta-learning, we evaluate two additional baselines under spatial CV with five seeds: ProtoNet (Snell et al., 2017) and Meta-Baseline (Chen et al., 2021) (Table 3, Figure 7). **ProtoNet is competitive with FOMAML and Reptile** at $K = 1$: it achieves the highest F_1 of *any* method in Copiapó (0.690 vs. Reptile 0.665) and Huasco (0.774 vs. Reptile 0.755), while trailing slightly in Elqui (0.436) and Limarí (0.485). That a metric-based meta-learner with *zero* gradient adaptation matches gradient-based meta-learners corroborates the interpretation of Raghu et al. (2020) that the gain comes from a transferable representation, not from the adaptation mechanism. In contrast, **Meta-Baseline under-performs** all other meta-learners (e.g., Copiapó $K = 1$ $F_1 = 0.461$), indicating that for tabular landslide features a frozen concat-pre-trained encoder with a cosine head is insufficient—episodic training (ProtoNet) or gradient adaptation (Reptile/FOMAML) is needed. Across all four target basins, at least one meta-learning method beats the Independent baseline at $K = 1$, confirming the advantage is robust to the choice of meta-learner.

5.11. Variogram validation of spatial cross-validation

To verify that the K-means spatial folds (Section 4.3) achieve genuine spatial separation rather than merely splitting autocorrelated pixels, we estimate the empirical variogram of Independent-model residuals for each target basin and fit a spherical model to extract the autocorrelation range (Figure A.16). The fitted ranges are 39.3 km (Copiapó), 58.5 km (Huasco), 20.9 km (Elqui), and 15.6 km (Limarí). Comparing these to the median distance between fold centroids at $K_{\text{folds}} = 5$ (70.0, 55.7, 43.6, and 54.2 km respectively), the fold separation *exceeds* the autocorrelation range in three of four basins; only Huasco is marginal

(−2.8 km), reflecting its larger spatial extent and inventory. Re-running the benchmark at $K_{\text{folds}} \in \{5, 10, 20\}$ confirms that the FOMAML advantage over Independent persists across all fold granularities (per-target mean lift of +0.4 to +2.6 pp, averaged over $K \in \{5, 10\}$ shots and three seeds; Figure A.18), demonstrating that the reported advantage is not an artifact of the $K_{\text{folds}} = 5$ choice. We adopt $K_{\text{folds}} = 5$ as the primary protocol because it balances spatial decorrelation against fold-wise sample sufficiency in the smaller basins.

5.12. Spatial CNN backbone: spatial context substantially improves accuracy

To quantify how much the point-wise feature representation costs in predictive accuracy, we re-run the four primary methods (Independent, Fine-tune, Reptile, FOMAML) with a small spatial CNN backbone consuming 32×32 patches over the 25 gridded feature channels (Section 4.2) on the largest target basin, Huasco (Figure A.17). **The CNN backbone outperforms the MLP by 6–11 pp F_1 across all methods and K values**. At $K = 1$, Independent CNN reaches $F_1 = 0.78$, already exceeding the MLP's best ($F_1 = 0.76$ for Reptile); FOMAML-CNN reaches 0.82. At $K = 20$, all CNN variants reach $F_1 \geq 0.93$ compared to ~ 0.84 for the MLP. The meta-learning lift over Independent is smaller in the CNN regime (+4.2 pp at $K = 1$ for FOMAML-CNN vs. +6.8 pp for FOMAML-MLP, with the two intervals overlapping), indicating that spatial context absorbs part of the prediction signal that meta-learning's prior captures in the point-wise regime. Two implications follow, and one competing interpretation must be stated. First, the main-benchmark MLP results report a *conservative lower bound* on achievable cross-basin transfer accuracy; the systematic +5.5 pp meta-learning lift remains valid as a methodological finding but is not the operational ceiling. Second, the framework is mechanically applicable to a spatial CNN backbone, and the meta-learning ordering over Independent is preserved in that regime. However, the shrinking lift also admits the alternative reading that *the meta-learning advantage is largest precisely where the backbone is weakest*—i.e., that a stronger spatial representation absorbs the transferable signal the prior would otherwise contribute—and a single-basin demonstration cannot distinguish these interpretations. A multi-target CNN/U-Net meta-learning evaluation is required to resolve them and is left for follow-up work; until then, the Huasco experiment should be read as a proof of mechanism, not as evidence that the K-shot advantage carries unchanged into stronger architectures.

6. Discussion

6.1. Reptile vs. FOMAML: comparable performance, computational trade-offs

Reptile and FOMAML achieve statistically comparable F_1 scores across both random and spatial-CV protocols, with bootstrap 95% confidence intervals that overlap in the majority of target-K cells (Table 2). At $K = 1$ under spatial CV, Reptile nominally leads in all four targets—Copiapó (0.665 vs. FOMAML 0.663, +0.2 pp), Huasco (0.755 vs. 0.745, +1.0 pp), Elqui (0.476 vs. 0.472, +0.4 pp), and Limarí (0.528 vs. 0.514, +1.4 pp)—while FOMAML leads at $K = 5$ in Copiapó (0.745 vs. 0.724); none of these inter-method differences exceed the bootstrap standard error. This convergence is consistent with Raghu et al. (2020), who argue that the principal benefit of MAML-family methods is rapid feature reuse, not the

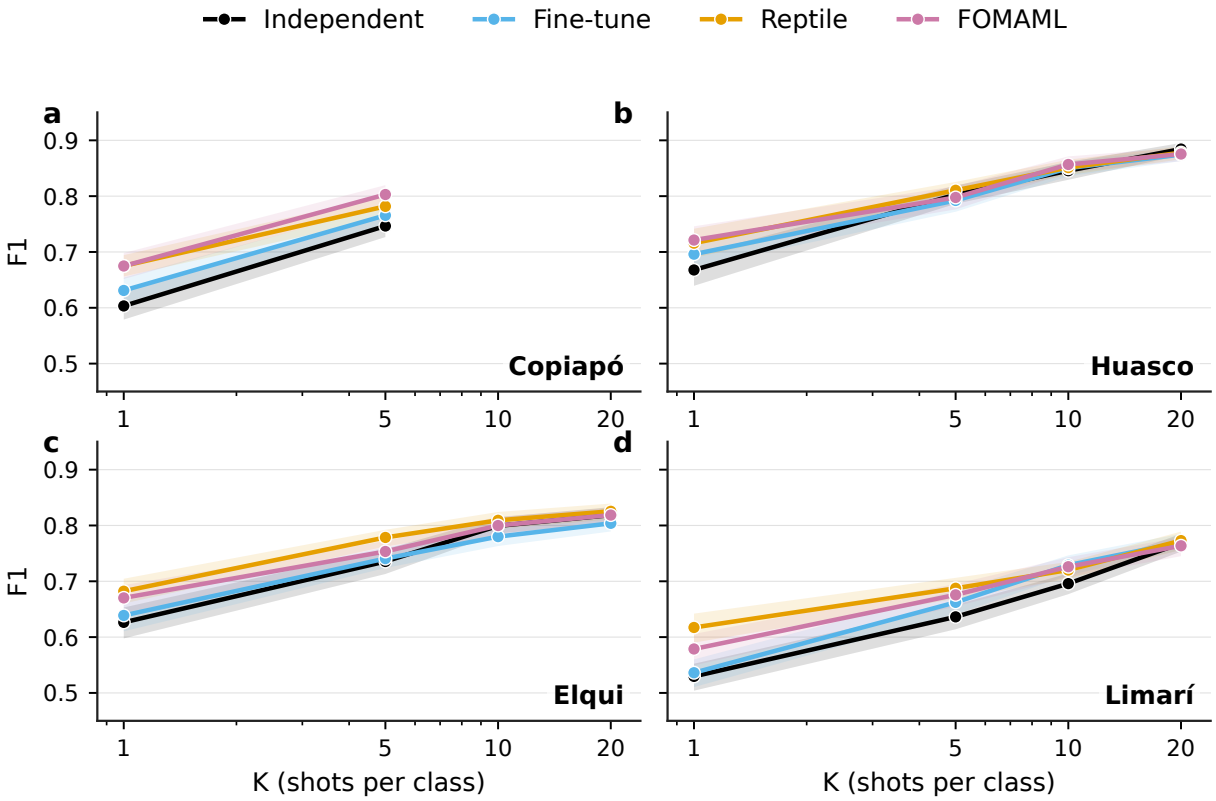


Figure 2: F_1 score versus number of support shots K under *random* cross-validation for the four target basins. Lines: mean across 40 episodes × 3 seeds; shaded bands: bootstrap 95% confidence intervals. Meta-learning (Reptile or FOMAML) outperforms Independent at $K = 1$ in all four basins; Reptile leads at $K = 1$ in Elqui and Limarí, FOMAML leads in Huasco, and the two are within overlapping CIs in Copiapó. The advantage between meta-learning and Independent narrows to < 1 pp at $K = 20$.

second-order curvature term that distinguishes the two algorithms. Practically, Reptile's update rule (average-of-inner-displacements) is computationally cheaper than FOMAML's (requires a query-loss forward pass through the inner-adapted model per outer step), so we recommend Reptile as the default meta-learning algorithm for tabular landslide susceptibility transfer; FOMAML is a statistically equivalent alternative whose only nominal edge in our benchmark appears at $K = 5$ in Copiapó (hyper-arid, very small inventory).

This near-equivalence extends to a third meta-learner: ProtoNet (Snell et al., 2017), a metric-based method that performs *no* gradient adaptation at test time, matches or exceeds the gradient-based methods at $K = 1$ (Section 5.10, Table 3). The convergence of three structurally distinct meta-learners—averaged-displacement (Reptile), query-gradient (FOMAML), and nearest-prototype (ProtoNet)—onto similar performance strongly supports the Raghu et al. (2020) thesis that the few-shot advantage is driven by a transferable feature representation rather than by any particular adaptation rule. The failure of Meta-Baseline (Chen et al., 2021), which freezes a concat-pre-trained encoder, indicates that this representation must be learned *episodically* (or refined by gradient adaptation): a represen-

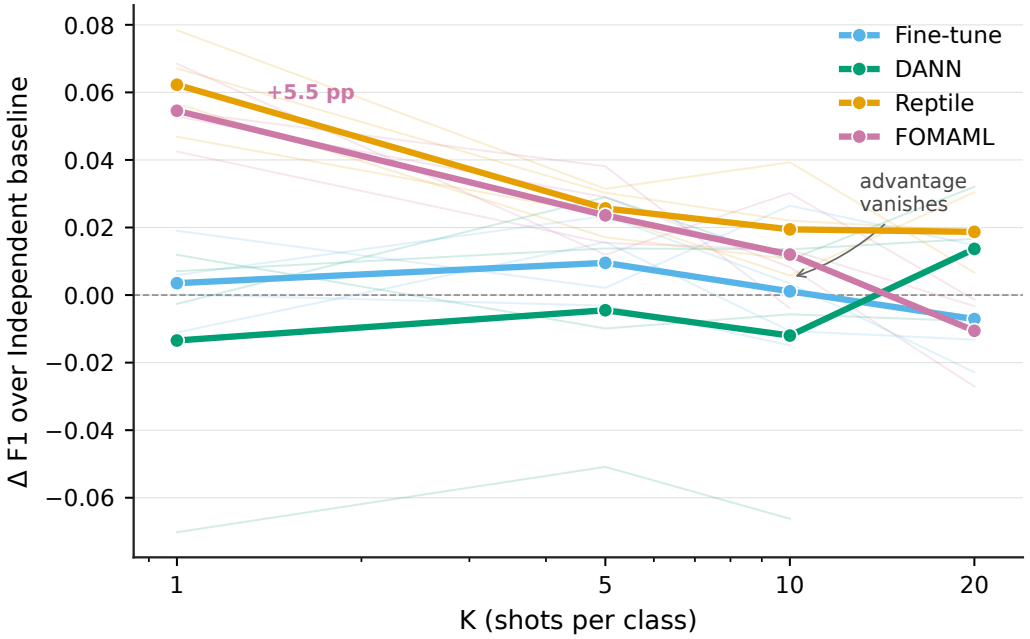


Figure 3: Mean F_1 advantage over the Independent baseline (averaged across the four target basins) under *spatial* cross-validation. FOMAML achieves +5.5 pp at $K = 1$, decaying monotonically to −1.1 pp at $K = 20$ (over the three basins where $K = 20$ is feasible); Reptile decays from +6.2 to +1.9 pp. DANN's advantage is negative at $K \leq 10$ (−0.5 to −1.4 pp) and +1.4 pp at $K = 20$. Thin lines: per-target advantages; thick lines: cross-target means.

tation optimized only for source classification does not transfer as well as one optimized for rapid task adaptation.

6.2. Why DANN fails in the smallest target basin

DANN does not improve over Independent on average. The largest point-estimate gap is in Copiapó, the smallest target basin in our experiment ($N = 19$ landslide events), where DANN trails Independent by 5–7 pp F_1 across all K values (Table 2); we caution, however, that this basin's nine spatial partitions do not resolve the gap as significant under the cluster-level paired test (−7.0 pp [−15.9, +1.5]; Table 4), so the Copiapó degradation is a suggestive pattern rather than an established effect. In the other three target basins (Huasco, Elqui, Limarí), DANN is roughly neutral (cluster-paired CIs all include zero). What is robust across the analysis is the *absence* of any DANN advantage: unlike meta-learning, DANN is significant in no target. This contradicts the narrative in much of the DA literature, where DANN is presented as a strong default baseline, and demands explanation. We propose two complementary mechanisms.

First, *adversarial invariance requires sufficient domain diversity.* With only $N = 10$ source basins, the domain classifier easily exploits low-level features such as climate (basin-mean bio_12 ranges from $\sim$ 1 mm/yr in the Atacama sources to $\sim$ 1,800 mm/yr in the humid Patagonian source Bueno-Puelo) or lithology (volcanic vs. sedimentary). The Gra-

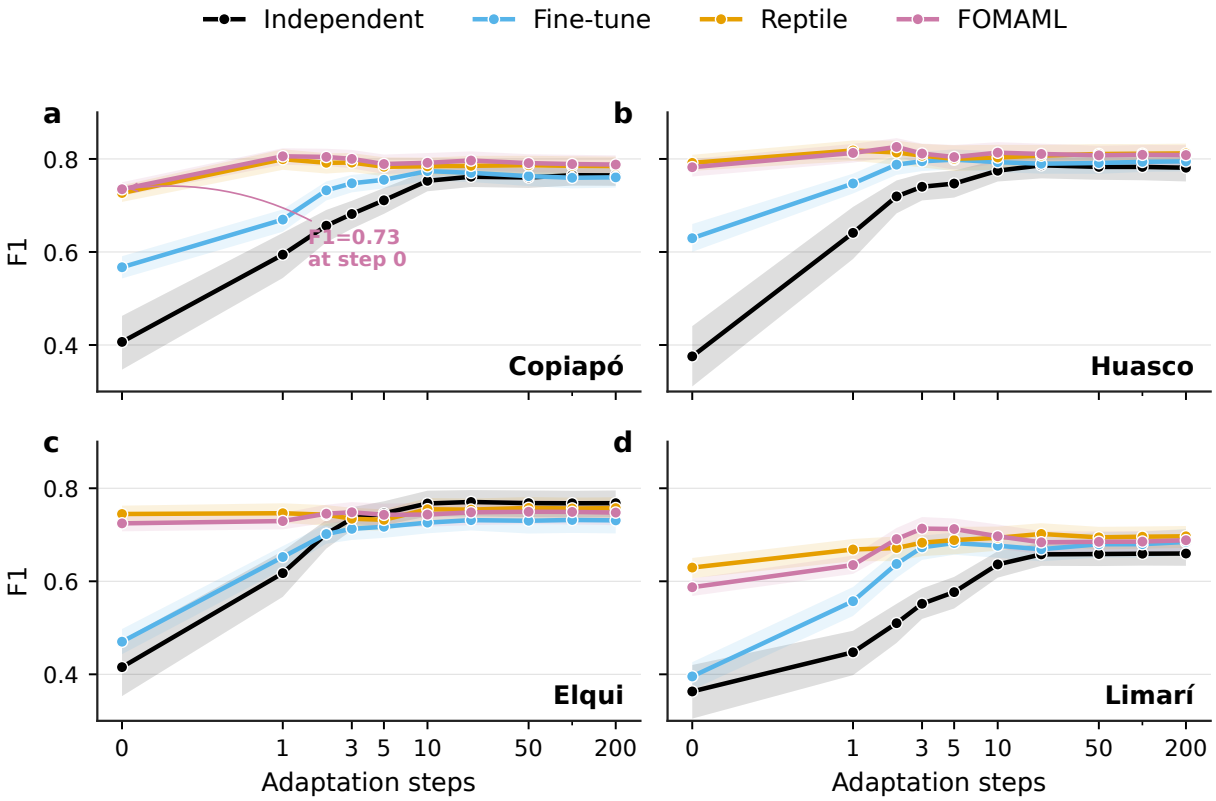


Figure 4: Adaptation efficiency at $K = 5$ across target basins. F_1 as a function of inner Adam gradient steps applied to the support set, evaluated on held-out queries. FOMAML reaches $F_1 \approx 0.78$ with *zero* gradient steps in Copiapó, indicating the meta-learned initialization is already discriminative. Reptile peaks in 3–5 steps; Independent requires >100 steps to converge.

dient Reversal Layer encourages the feature extractor to discard these signals, but they are also potentially predictive of landslide susceptibility (e.g., precipitation-driven debris flows correlate with bio_12). The result is a representation that is more domain-invariant but less class-discriminative—a form of *negative transfer* where alignment removes signal.

Second, *the K-shot adaptation step is poorly matched to DANN's design philosophy.* DANN was designed for the unsupervised setting in which target labels are entirely unavailable. Re-purposing the trained DANN as a starting point for a small fine-tune on K labels conflates two mechanisms (adversarial invariance + supervised fine-tune) that are not jointly optimized. Two alternative formulations—MMD-based DA (Long et al., 2015) and source-target joint embeddings—are reasonable next baselines but are outside our current scope.

6.3. Spatial heterogeneity in Elqui: a hard transfer case

Among target basins, Elqui exhibits the largest drop in F_1 between random and spatial CV protocols (27 pp at $K = 10$; Table 2 vs. Figure A.8). All methods, including FOMAML,

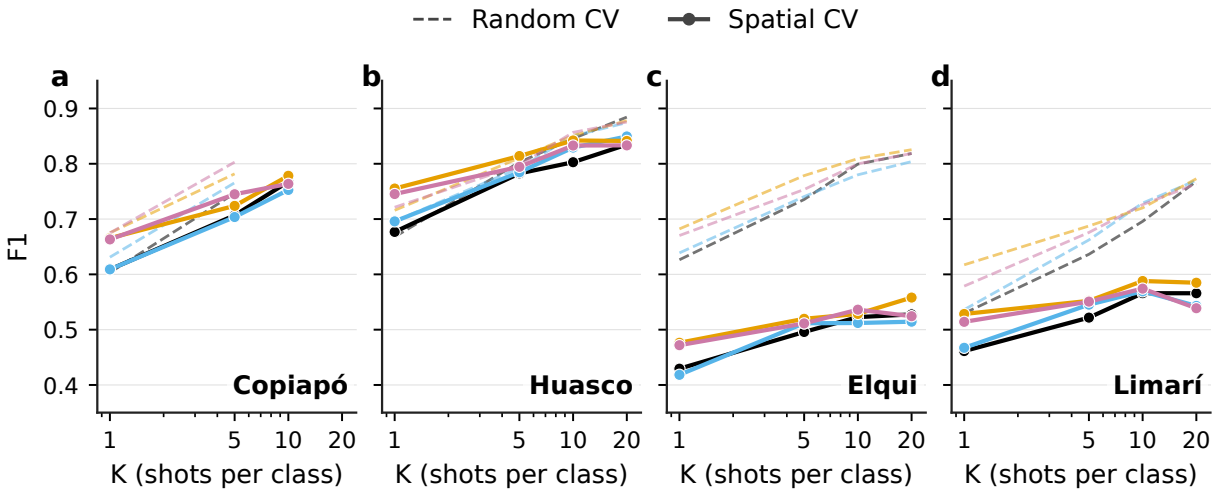


Figure 5: F_1 versus K under random (dashed) and spatial (solid) cross-validation. Spatial CV partitions basins into 5 K-means clusters on (x, y) coordinates, ensuring support and query are spatially disjoint. Drops between protocols: 1 pp (Huasco) to 27 pp (Elqui), but the relative ranking of methods is preserved.

are constrained to $F_1 \approx 0.50$ under spatial CV at any K. This indicates that intra-basin heterogeneity, not the cross-basin meta-learning capability, is the dominant factor in Elqui.

Two hypotheses are consistent with the data. (i) **Orographic structure**: Elqui has a strong west-east elevation gradient (sea level to $\sim$ 5,000 m), with the cordillera dominated by Mesozoic plutonic rocks (granitoids), the precordillera by Cenozoic volcanics, and the coast by Quaternary alluvial deposits. Each lithological unit produces distinct landslide styles (debris flows in the alluvium, rotational slides in the granitoids), and our 11-feature representation may not capture these axes adequately. (ii) **Trigger heterogeneity**: the SERNAGEOMIN *catastro* for Elqui aggregates events triggered by the 2015 Atacama floods, the 1997 El Niño, and the 1922 Vallenar earthquake, three regimes with distinct preparatory factors. A meta-model trained on Maule 2010 (seismic) and Chañaral (rainfall) sources cannot easily recover this trigger-specific structure from the support set alone.

Future work should address Elqui-style heterogeneity by (a) richer spatial features (e.g., multi-temporal Sentinel-1 SAR coherence), (b) trigger-aware meta-learning, or (c) fold-specific stratification. None of these are straightforward extensions, and we report Elqui as a transparent failure mode rather than attempting to mask it.

6.4. Climate and transfer: a preliminary null

The absence of a detectable correlation between source–target precipitation distance and transfer advantage (Section 5.8) is a useful practical hint, but it is a preliminary null from an underpowered test (four targets, one climatic variable), not established climate invariance. Read at that strength, it is *consistent with* the interpretation that the meta-learned prior captures generic geomorphic "susceptibility patterns" (e.g., concave hillslopes, high MRVBF, low landform variability) rather than climate-specific signals—an interpretation

Table 2: F_1 score (mean) with bootstrap 95% confidence intervals for each target basin × shots-per-class K × method, under spatial cross-validation. Best per row in **bold**. Copiapó $K = 20$ is not reported because the basin's inventory of $N = 19$ events is too small to support $K = 20$ positives in every spatial-CV fold. Valid fold × seed combinations are basin-dependent (Elqui 15/15, Limarí 12/15, Copiapó 9/15, Huasco 6/15; Section 4.3); total realized K-shot adaptations: 15,900. Source: `results/spatial_benchmark/summary.csv`.

Target	K	Independent	Fine-tune	DANN	Reptile	FOMAML
Copiapó	1	0.609 [.58,.64]	0.609 [.57,.64]	0.539 [.50,.58]	**0.665 [.63,.70]**	0.663 [.63,.70]
	5	0.707 [.68,.74]	0.704 [.67,.74]	0.656 [.62,.69]	0.724 [.69,.76]	**0.745 [.71,.78]**
	10	0.768 [.74,.80]	0.753 [.72,.78]	0.701 [.67,.73]	**0.778 [.75,.81]**	0.764 [.73,.80]
Huasco	1	0.677 [.64,.71]	0.696 [.66,.73]	0.689 [.66,.72]	**0.755 [.73,.78]**	0.745 [.72,.77]
	5	0.783 [.77,.80]	0.785 [.77,.80]	0.773 [.76,.79]	**0.814 [.80,.83]**	0.794 [.78,.81]
	10	0.803 [.79,.82]	0.829 [.82,.84]	0.797 [.78,.81]	**0.842 [.83,.86]**	0.833 [.82,.85]
	20	0.835 [.82,.85]	**0.849 [.84,.86]**	0.827 [.81,.84]	0.841 [.83,.85]	0.833 [.82,.85]
Elqui	1	0.429 [.40,.46]	0.418 [.39,.45]	0.437 [.41,.47]	**0.476 [.45,.51]**	0.472 [.44,.51]
	5	0.496 [.46,.53]	0.512 [.48,.54]	0.510 [.48,.54]	**0.520 [.49,.56]**	0.511 [.48,.54]
	10	0.523 [.49,.56]	0.512 [.48,.55]	0.536 [.50,.57]	0.528 [.49,.57]	**0.536 [.50,.57]**
	20	0.528 [.49,.57]	0.514 [.48,.55]	0.544 [.51,.59]	**0.558 [.52,.60]**	0.524 [.48,.57]
Limarí	1	0.461 [.43,.49]	0.467 [.44,.50]	0.459 [.43,.49]	**0.528 [.50,.56]**	0.514 [.49,.54]
	5	0.522 [.49,.55]	0.545 [.51,.58]	0.551 [.52,.58]	**0.552 [.52,.58]**	0.551 [.51,.58]
	10	0.566 [.53,.60]	0.569 [.53,.60]	0.576 [.55,.60]	**0.588 [.55,.62]**	0.574 [.54,.61]
	20	0.566 [.54,.59]	0.543 [.50,.58]	**0.598 [.57,.62]**	0.585 [.55,.61]	0.539 [.51,.57]

independently supported by the CFS selection, in which DEM morphometry dominates and precipitation enters as a single feature. The tentative design implication is that source-pool *breadth* across climate regimes is a reasonable default and that climate matching need not be the first-order source-selection criterion; confirming or refuting a genuine climate dependence requires more targets and a multivariate environmental distance, which the released benchmark is designed to enable.

6.5. Implications for remote-sensing-driven susceptibility mapping

Three findings carry methodological weight specifically for the remote-sensing-based geohazard community. *First*, the meta-learned prior is recovered from an entirely open, globally-accessible feature stack—MERIT DEM derivatives, WorldClim climatology and coarse geology—without proprietary inputs, optical composites or in-situ measurements. This makes the framework directly replicable in any region with global-DEM coverage (effectively everywhere on Earth). The Chilean focus is illustrative, not restrictive. *Second*, CFS-selected features (Section 3.5) reveal that DEM-derived terrain and GLCM-texture descriptors dominate the discriminative signal; the sole non-DEM feature retained is WorldClim annual precipitation (bio_12), while hydrology and geology descriptors are not selected. This has practical consequences: a 30 m DEM of moderate quality (MERIT, Copernicus GLO-30, or AW3D30) is the binding constraint on framework deployment, not optical-sensor availability. Countries without dense optical archives still have global DEM access. *Third*, the meta-learning advantage is largest at $K = 1$, which corresponds to one verified mass-

Table 3: F_1 score (mean) with bootstrap 95% confidence intervals for the three additional baselines (ProtoNet, Meta-Baseline, CDAN) under spatial cross-validation with five seeds. Best per row in **bold**. For comparison against the primary methods (Independent, Fine-tune, DANN, Reptile, FOMAML) see Table 2. Copiapó $K = 20$ omitted ($N = 19$, see Table 2). Source: `results/extended_baselines/summary.csv`.

Target	K	ProtoNet	Meta-Baseline	CDAN
Copiapó	1	**0.690 [.66,.71]**	0.461 [.43,.49]	0.573 [.54,.60]
	5	**0.727 [.70,.75]**	0.494 [.47,.52]	0.673 [.65,.70]
	10	**0.774 [.75,.80]**	0.531 [.50,.56]	0.716 [.69,.74]
Huasco	1	**0.774 [.75,.80]**	0.681 [.66,.70]	0.707 [.68,.73]
	5	**0.836 [.83,.84]**	0.766 [.76,.77]	0.768 [.75,.78]
	10	**0.840 [.83,.85]**	0.772 [.77,.78]	0.807 [.80,.82]
	20	**0.840 [.84,.85]**	0.771 [.77,.77]	0.833 [.82,.84]
Elqui	1	0.436 [.41,.46]	0.394 [.37,.42]	**0.440 [.42,.46]**
	5	**0.526 [.50,.55]**	0.479 [.46,.50]	0.518 [.49,.54]
	10	0.528 [.50,.55]	0.498 [.48,.52]	**0.543 [.52,.57]**
	20	0.520 [.50,.54]	0.511 [.49,.53]	**0.540 [.51,.57]**
Limarí	1	**0.485 [.46,.51]**	0.412 [.39,.43]	0.433 [.41,.45]
	5	0.536 [.51,.56]	0.510 [.49,.53]	**0.552 [.53,.57]**
	10	0.543 [.52,.56]	0.543 [.53,.56]	**0.596 [.58,.61]**
	20	0.566 [.55,.58]	0.580 [.57,.59]	**0.607 [.59,.62]**

movement footprint visible in high-resolution Sentinel-2 or Planet imagery. This aligns with operational reality: a remote-sensing analyst can typically validate a single landslide via visual interpretation in < 10 minutes, whereas mapping the $K \geq 50$ events required by per-basin training takes weeks. Meta-learning thus reduces the human-in-the-loop cost from *inventory construction* (weeks) to *seed-event validation* (minutes), a regime where rapid operational deployment becomes feasible. In the Chilean institutional context, the resulting maps (Figure 6) are directly consumable by the agencies that motivated this study: SERNAGEOMIN, which maintains the national *catastro de remociones en masa* used here as the label source, and SENAPRED, whose regional risk-management plans require basin-scale susceptibility layers as inputs for land-use and emergency planning. Because a $K = 1$ adaptation needs a single verified event, the framework allows a first-pass susceptibility layer for an unmapped basin to be produced within hours of a triggering event, rather than after a months-long inventory campaign—with the explicit caveat that in heterogeneous basins such as Elqui the resulting layer inherits the low discrimination quantified in Table 2 and should be treated as a screening product, not a planning-grade map. A natural extension is to incorporate multi-temporal Sentinel-1 InSAR coherence and amplitude features (Poggi et al., 2026; Biggs et al., 2026; Elliott et al., 2026) to capture surface-deformation precursors that static optical-DEM stacks cannot, an avenue actively explored for global InSAR landslide monitoring (Guo et al., 2026) and likely to amplify the meta-learning advantage by injecting time-varying signals into the support set.

Table 4: Cluster-level paired comparison of each method against the Independent baseline at $K = 1$ under spatial cross-validation. The paired unit is the (seed, fold) spatial partition (n_c per target), within which all methods evaluate on the same held-out query cluster; the paired difference is the per-cluster mean-F_1 gap. Intervals are 95% hierarchical-bootstrap CIs (resampling clusters, then episodes; 2,000 resamples). *CI excludes zero. The last columns give the number of clusters with a positive difference and the two-sided Wilcoxon signed-rank p-value on the cluster differences. Meta-learning (Reptile/FOMAML) is significant in Huasco, Elqui and Limarí but not in Copiapó ($N = 19$, too few clusters); Fine-tune and DANN are significant in no target. Source: `results/stats/cluster_paired_vs_independent.csv`.

Target	**Method**	ΔF_1 **vs Indep. (pp) [95% CI]**	**Clusters** > 0	**Wilcoxon** p
Copiapó ($n_c = 9$)	Reptile	+5.7 [-0.1, +12.9]	7/9	0.164
	FOMAML	+5.4 [-0.6, +12.3]	7/9	0.098
	Fine-tune	+0.0 [-5.2, +6.1]	5/9	1.000
	DANN	−7.0 [-15.9, +1.5]	3/9	0.250
Huasco ($n_c = 6$)	Reptile	+7.8 [+4.8, +11.6]*	6/6	0.031
	FOMAML	+6.8 [+2.4, +10.9]*	5/6	0.062
	Fine-tune	+1.9 [-2.0, +5.6]	4/6	0.438
	DANN	+1.2 [-2.7, +5.2]	3/6	0.688
Elqui ($n_c = 15$)	Reptile	+4.7 [+2.6, +6.9]*	13/15	0.001
	FOMAML	+4.2 [+1.5, +7.3]*	12/15	0.012
	Fine-tune	−1.1 [-3.9, +1.3]	6/15	0.561
	DANN	+0.7 [-4.6, +6.9]	5/15	0.639
Limarí ($n_c = 12$)	Reptile	+6.7 [+3.4, +9.8]*	10/12	0.007
	FOMAML	+5.3 [+1.5, +8.9]*	10/12	0.034
	Fine-tune	+0.6 [-3.9, +5.0]	6/12	0.970
	DANN	−0.3 [-4.6, +3.7]	7/12	0.970

6.6. Limitations

We acknowledge six limitations.

Source domain count. With 10 source basins, the meta-learning regime is at the lower end of typical few-shot benchmarks (Omniglot uses 1,623 classes, miniImageNet uses 64 source classes). Adding more basins would likely increase the meta-learning advantage. Adding basins from neighboring countries (Peru, Argentina, Bolivia, Ecuador) is a natural next step.

MLP backbone. The main benchmark uses a 3-layer MLP on point-wise tabular features. This is intentionally conservative: it ensures interpretability, avoids large-scale GPU requirements, and maps cleanly to the practical workflow of basin managers. The U-Net-style CNN baseline of Section 5.12 quantifies the cost of this choice on Huasco: the spatial CNN improves F_1 by 6–11 pp across methods and K values, and the meta-learning lift is partially absorbed by the spatial representation. The MLP results should therefore be read as a conservative lower bound on achievable accuracy. A multi-target U-Net meta-learning evaluation, including richer patch sizes and attention-based architectures, is the natural follow-up.

Table 5: Sensitivity of the $K = 1$ meta-learning advantage to non-landslide (negative) sample selection, under spatial cross-validation (mean F_1 over three seeds). Target negatives are re-generated by random sampling outside exclusion buffers of 500 m, 1 km and 2 km, and by a slope-matched strategy whose negatives reproduce the positives' slope distribution (removing the steep-vs-flat confound). The meta-learned prior (Reptile) reuses the unchanged source meta-model. Features are re-extracted from rasters, so absolute F_1 differs slightly from Table 2; the comparison across conditions is internally consistent. Absolute F_1 shifts with the sampling choice (consistent with Abdelkader et al. 2025), but the meta-learning *lift* stays positive throughout (range +3.0 to +8.1 pp) and *increases* under the hardest slope-matched condition. Source: `results/stats/negatives_sensitivity.csv`.

Target	**Negative-sampling strategy**	**Indep.** F_1	**Reptile** F_1	**Lift (pp)**
Huasco ($N = 278$)	Random, 500 m buffer (baseline)	0.734	0.775	+4.1
	Random, 1 km buffer	0.713	0.750	+3.8
	Random, 2 km buffer	0.733	0.763	+3.0
	Slope-matched	0.654	0.709	+5.5
Copiapó ($N = 19$)	Random, 500 m buffer (baseline)	0.700	0.730	+3.0
	Random, 1 km buffer	0.593	0.647	+5.4
	Random, 2 km buffer	0.585	0.666	+8.1

Static climate. Our climate features (WorldClim 2.1 bioclimatics) represent a 1970–2000 baseline. Climate-change-driven shifts in precipitation regimes are a known driver of changing landslide hazard but are absent from our model. Coupling with CMIP6 projections is straightforward in principle.

Probability calibration and prevalence. Two coupled caveats govern how the outputs should be read. First, the models are evaluated on balanced 1:1 queries at a fixed 0.5 threshold; under the extreme class imbalance of real basins, absolute precision is far lower than the balanced F_1 suggests (Section 5.3, Figure A.9). Second, a model adapted on only $2K$ support samples has no calibration guarantee, so its raw sigmoid outputs are not probabilities in the frequentist sense—which is why the susceptibility maps (Figure 6) are shown as within-basin percentiles. Both caveats point to the same operational conclusion: the framework should be used as a *ranking/stratification* tool, evaluated with prevalence-independent metrics (ROC-AUC, average precision), rather than as a per-pixel binary classifier. Where a calibrated probability is required, post-hoc calibration on the K-shot support (Platt scaling or isotonic regression) is the natural next step, at the cost of consuming part of the already-scarce support budget.

Negatives generation. Eight of fourteen basins lack explicit negative samples, requiring us to generate them by random sampling within the basin polygon outside a 500 m buffer of positives. This introduces an implicit assumption—that all unlabeled pixels are negatives—which is statistically defensible (landslide footprints are sparse) but methodologically simplistic, and non-landslide sample selection is known to affect absolute susceptibility performance (Abdelkader et al., 2025). To test whether our conclusion depends on this choice, we re-ran the $K = 1$ spatial-CV benchmark on the two exemplar targets (Huasco, Copiapó) under three exclusion-buffer widths (500 m, 1 km, 2 km) and a slope-matched strategy

that reproduces the positives' slope distribution (Table 5). As expected, absolute F_1 shifts with the sampling scheme, but the meta-learning *lift* over Independent remains positive in every condition (range +3.0 to +8.1 pp) and, tellingly, *grows* under the slope-matched scheme (+5.5 pp in Huasco): when the easy "steep-versus-flat" confound is removed, the meta-learned prior helps more, not less, indicating the advantage is not an artifact of easy random negatives. Positive-Unlabeled (PU) learning remains a more rigorous alternative to the all-unlabeled-are-negative assumption.

Single-trigger framing. Our inventory mixes seismic and rainfall-triggered events without explicit modeling of trigger type. Trigger-aware sampling and feature design would likely improve discrimination, particularly in mixed-trigger basins like Elqui.

6.7. Practical recommendations for new-basin susceptibility modeling

For a researcher or basin manager facing a *newly-studied* watershed with fewer than 10 documented landslide events, our findings support the following protocol:

1. **Identify a meta-training pool** of at least 4 source basins from a diverse range of hydro-climatic regimes. Climate matching does not appear to be the first-order source-selection concern, though the evidence is preliminary (Section 5.8).
2. **Use Reptile as the default meta-learner** for tabular feature representations: it is computationally cheaper than FOMAML and statistically equivalent in our benchmark (Section 6.1). FOMAML is a reasonable alternative in hyper-arid, very-small-inventory settings resembling Copiapó.
3. **Start from pooled logistic regression** (concat-source mode) as the operational default: it is competitive with meta-learning in peak F_1 at every K we tested, is simpler to train and audit, and requires no meta-training infrastructure (Section 5.9). Reach for the meta-learned initialization when its distinctive property matters—an initialization that specialises to a new basin in ≤ 3 gradient steps—or when the goal is to test whether a transferable prior exists across a basin ensemble.
4. **Avoid DANN** when source-domain count is small ($N \leq 10$). A simple concat-source fine-tune or independent training is at least as effective and substantially simpler.
5. **Validate with spatial cross-validation**, not random hold-out. Random CV inflates absolute F_1 by 5–25 pp depending on intra-basin spatial autocorrelation; the *relative* method ranking is preserved, but the *absolute* numbers reported to stakeholders should come from spatial CV.
6. **Adapt for ≤ 30 gradient steps**. Meta-learned initializations reach near-peak F_1 within 0–5 steps; further adaptation provides diminishing returns and risks over-fitting on the small support set.
7. **Document hard cases**. Basins with strong intra-basin heterogeneity (multiple lithologies, multiple triggers, strong elevation gradient) may resist transfer and require trigger-stratified or lithology-stratified analysis as a follow-up.

7. Conclusions

We present the first cross-basin application of first-order meta-learning (Reptile, FOMAML) to landslide susceptibility classification across 14 Chilean watersheds spanning an extreme Atacama-to-Patagonia climate gradient, evaluated with rigorous spatial cross-validation and complementary to the temporal single-site setting of Ma et al. (2025). With 70,200 K-shot model fits across methods, K-shot regimes, three to five seeds, and dual cross-validation protocols, we demonstrate that meta-learning (Reptile or FOMAML) outperforms an Independent baseline at the most extreme few-shot regime ($K = 1$) in 4 of 4 target basins under spatial cross-validation, with a mean FOMAML lift of +5.5 pp F_1 (range +4.3 to +6.8 pp; Reptile mean +6.2 pp, up to +7.8 pp). A cluster-level paired test that respects the spatial-fold structure confirms this advantage is statistically significant in three of the four targets (Huasco, Elqui, Limarí; Wilcoxon $p \leq 0.03$), and of similar magnitude but not resolvable in the smallest ($N = 19$ Copiapó, nine spatial partitions). Reptile and FOMAML achieve statistically comparable performance with overlapping bootstrap CIs in most target-K cells. The advantage decays with K and disappears by $K = 20$, defining the practical envelope of applicability.

Three findings carry methodological weight beyond the landslide application:

1. **A transferable geomorphic prior exists, and how it is extracted matters**. Within one architecture, episodic meta-training transfers across basins while naive pooled pretraining does not: Fine-tune gains ±1 pp over Independent and a frozen concat-pre-trained encoder (Meta-Baseline) fails outright, whereas three structurally distinct meta-learners (Reptile, FOMAML, ProtoNet) converge on similar gains. This is the paper's central claim—not predictive superiority, since a pooled-source logistic regression remains competitive in peak F_1 at every K we tested. What meta-learning uniquely provides is an initialization that specialises to a new basin in ≤ 3 gradient steps.
2. **Precipitation similarity shows no detectable relation to transfer success**. With four targets and a single-variable proxy the test is underpowered, so this is a preliminary null rather than established climate invariance; it is nonetheless consistent with a prior built on DEM morphometry, and suggests climate matching is not the first-order source-selection criterion.
3. **Domain-adversarial alignment does not help, and may hurt with very small targets**. DANN improves over Independent in no target, and its largest point-estimate deficit is in Copiapó ($N = 19$, 5–7 pp F_1), although that basin's nine spatial partitions do not resolve the deficit as significant. The robust statement is the absence of any adversarial advantage under a small source pool ($N = 10$).

Limitations include the use of a small tabular MLP backbone (vs. U-Net spatial models), static climate features, and a 10-source pool that is small by meta-learning standards. Future work should extend to U-Net spatial architectures, trigger-aware meta-learning to handle intra-basin heterogeneity (e.g., Elqui), and richer source pools spanning Andean countries.

For practitioners, the operational recommendation is deliberately modest: a pooled-source logistic regression is a legitimate default for basin-scale susceptibility screening, and the meta-learned initialization earns its added complexity in the narrow regime this benchmark delimits—one or two verified events, where a specialised model must be produced in minutes rather than after an inventory campaign. The released benchmark, datasets and evaluation protocol are intended to make that regime testable elsewhere in Latin America, where data-scarce basins outnumber data-rich ones and where rigorous spatial validation remains the exception rather than the rule.

CRediT author contribution statement

Francisco Parra: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Verónica Gil-Costa**: Methodology, Validation, Writing – review & editing, Supervision. **Carolina Bonacic**: Methodology, Validation, Writing – review & editing. **Mauricio Marín**: Conceptualization, Resources, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research was supported by the Postdoctoral Project DICYT 062619MC_POSTDOC, Universidad de Santiago de Chile. The funding source had no involvement in the study design; collection, analysis or interpretation of data; writing of the report; or the decision to submit the article for publication.

Acknowledgments

The authors thank SERNAGEOMIN (Servicio Nacional de Geología y Minería de Chile) for public access to the national *catastro de remociones en masa*, and Serey et al. (2019) for making the 2010 Maule earthquake landslide inventory publicly available. Map lines in this article delineate study areas and do not necessarily depict accepted national boundaries.

Data and code availability

- Code (pipeline scripts, figure-generation scripts, LaTeX source, aggregated benchmark summaries): https://github.com/franciscoparrao/paper4-meta-landslide.

- Per-basin ML-ready H5 datasets and CNN patch tensors are regenerated from public sources (SERNAGEOMIN *catastro de remociones en masa*; MERIT-DEM; WorldClim 2.1; SERNAGEOMIN geology; DGA Cuencas-BNA polygons) following the instructions in the repository README; a Zenodo deposit of the harmonised intermediate products is planned upon acceptance.

- SurtGIS toolkit, used for terrain and hydrology factor computation, is described in Parra (2026) and available at https://github.com/franciscoparrao/surtgis.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Code (Anthropic) to assist with prose drafting, code refactoring, statistical-result tabulation, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Supplementary figures

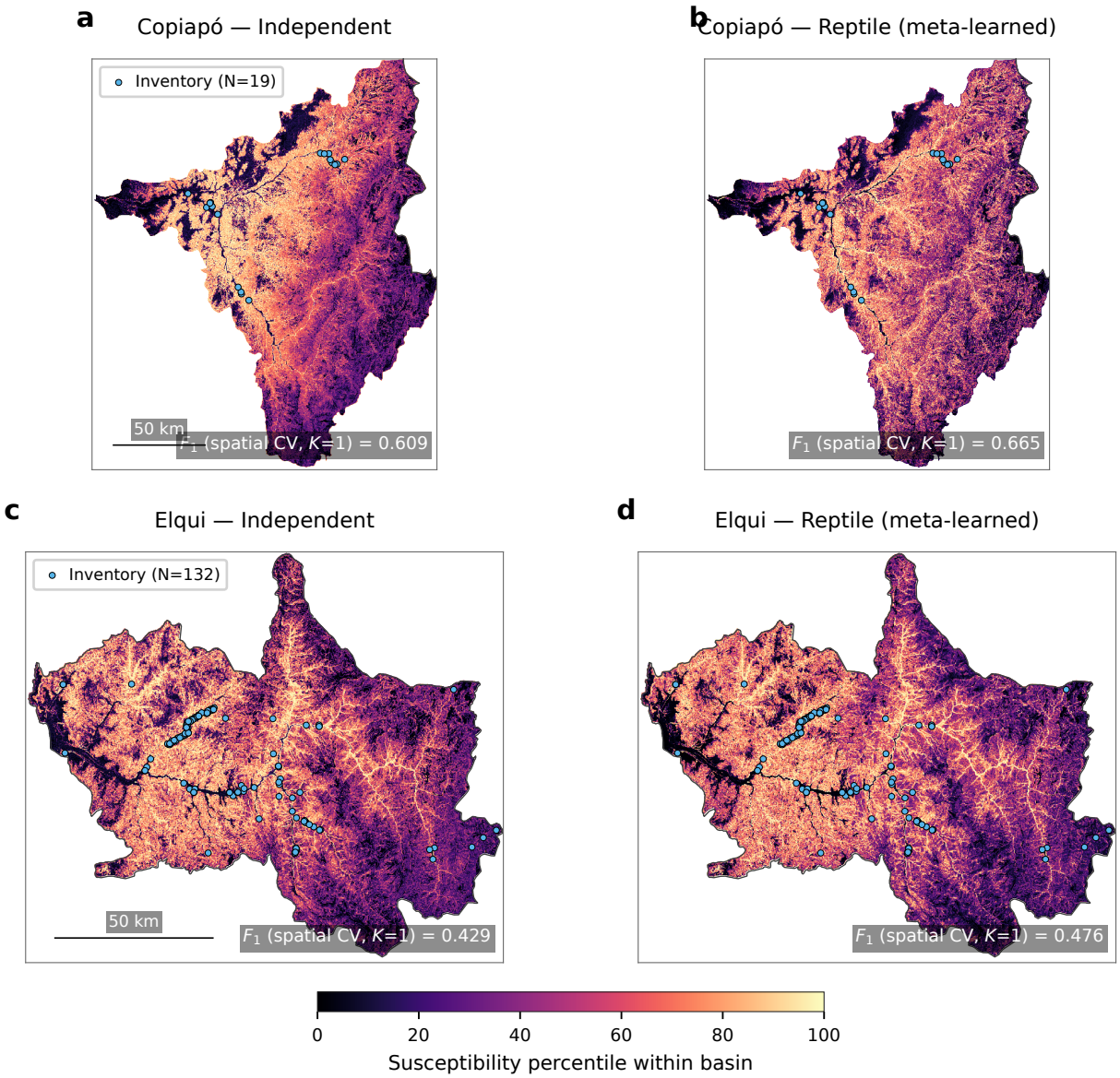


Figure 6: Full-basin susceptibility maps from $K = 1$-adapted models under the spatial-CV protocol, for Copiapó (top; smallest inventory, $N = 19$, largest AUC and average-precision gain, Section 5.3) and Elqui (bottom; a hard low-signal basin). Columns: Independent baseline (**a**, **c**) versus the meta-learned Reptile initialization (**b**, **d**), each the mean output over 3 seeds × 20 support draws, shown as within-basin susceptibility percentiles (raw $K = 1$ outputs are uncalibrated; Section 6.6). Blue points: SERNAGEOMIN inventory events. The Reptile surface sharpens the ridge–valley contrast and lifts the inventory-bearing western tributaries of Copiapó into higher percentiles, consistent with its +5.7 pp F_1 edge; in Elqui the diffuse pattern and scattered inventory make the low $F_1 \approx 0.48$ (Table 2) visible on the ground. The 50 km scale bar and per-panel spatial-CV F_1 ($K = 1$) are annotated.

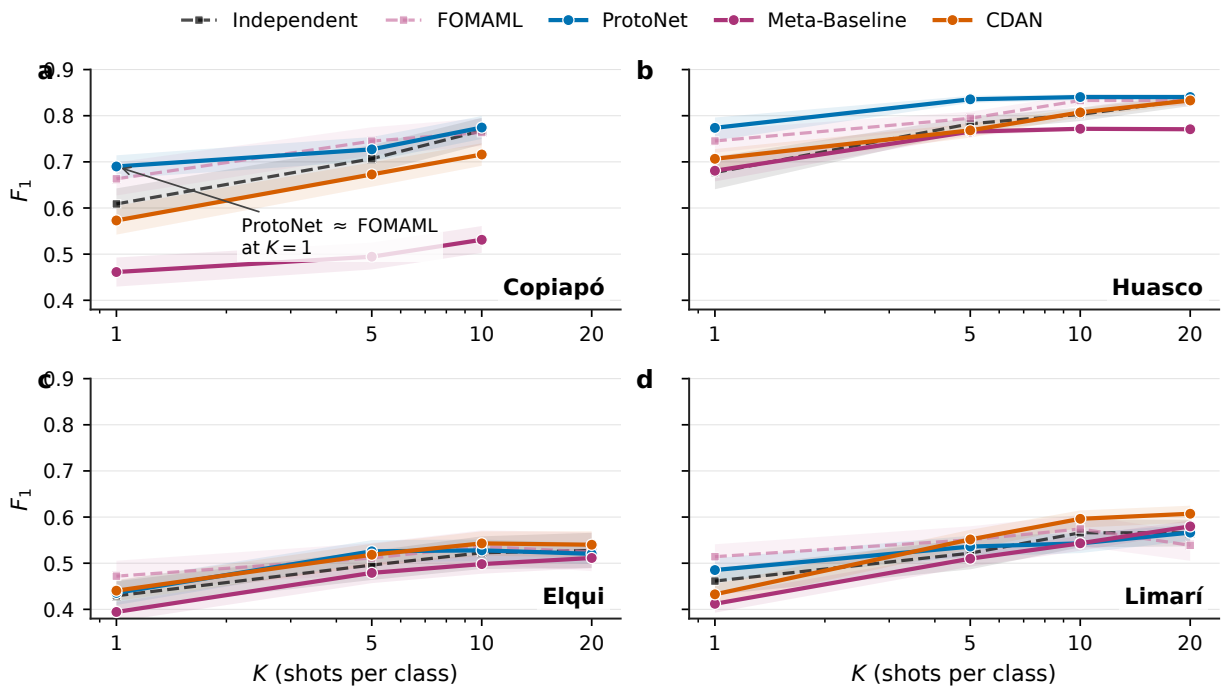


Figure 7: Extended baselines under spatial cross-validation (five seeds): ProtoNet, Meta-Baseline, and CDAN (solid lines) compared against the reference Independent baseline and FOMAML (dashed). ProtoNet, a metric-based meta-learner requiring no gradient adaptation, matches or exceeds FOMAML at $K = 1$ in Copiapó and Huasco, confirming the few-shot advantage is robust across meta-learning families. Meta-Baseline (frozen concat-pre-trained encoder) consistently lags. Shaded bands are bootstrap 95% confidence intervals.

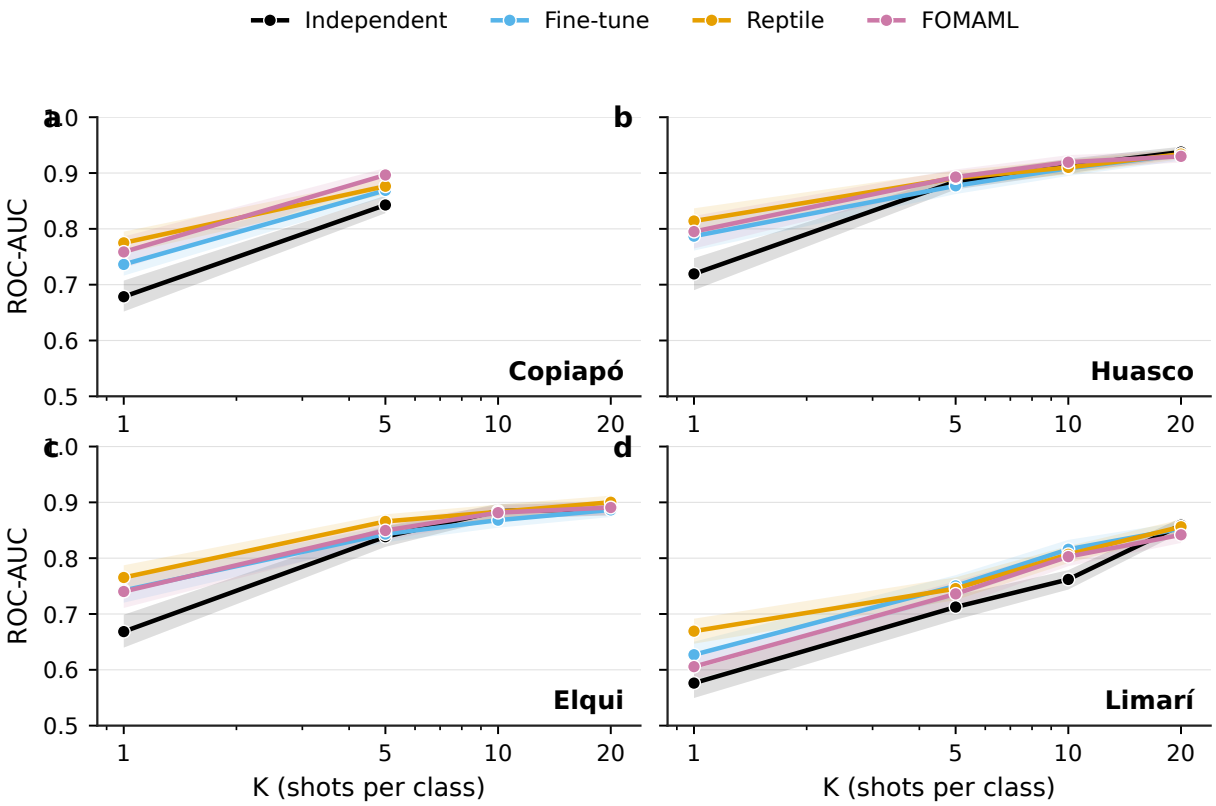


Figure A.8: ROC-AUC versus K under random cross-validation. Pattern consistent with F_1 results in Figure 2.

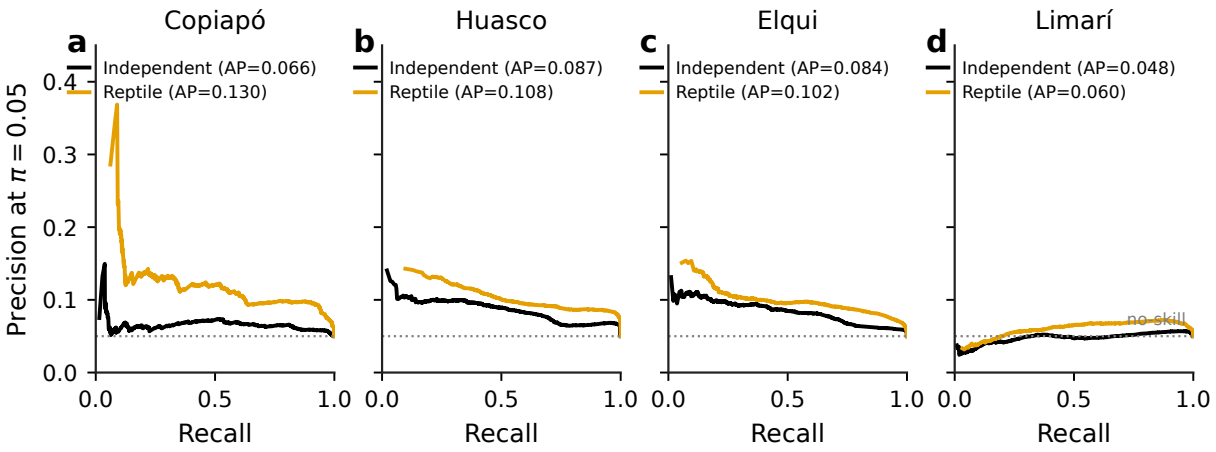


Figure A.9: Precision–recall re-weighted to a realistic landslide prevalence ($\pi = 0.05$) for the pooled $K = 1$ spatial-CV queries (Independent vs. Reptile), per target basin. Curves are obtained from the balanced operating characteristics via $\text{precision}(\pi) = \pi\,\text{TPR}/[\pi\,\text{TPR} + (1-\pi)\,\text{FPR}]$; the dotted line marks the no-skill baseline (precision $= \pi$). Absolute precision is low at operational prevalence—most flagged pixels are false positives, a property masked by the balanced F_1—yet the meta-learned Reptile prior dominates Independent in average precision (AP, legend) in all four targets, confirming the $K = 1$ advantage is prevalence-robust (Section 5.3).

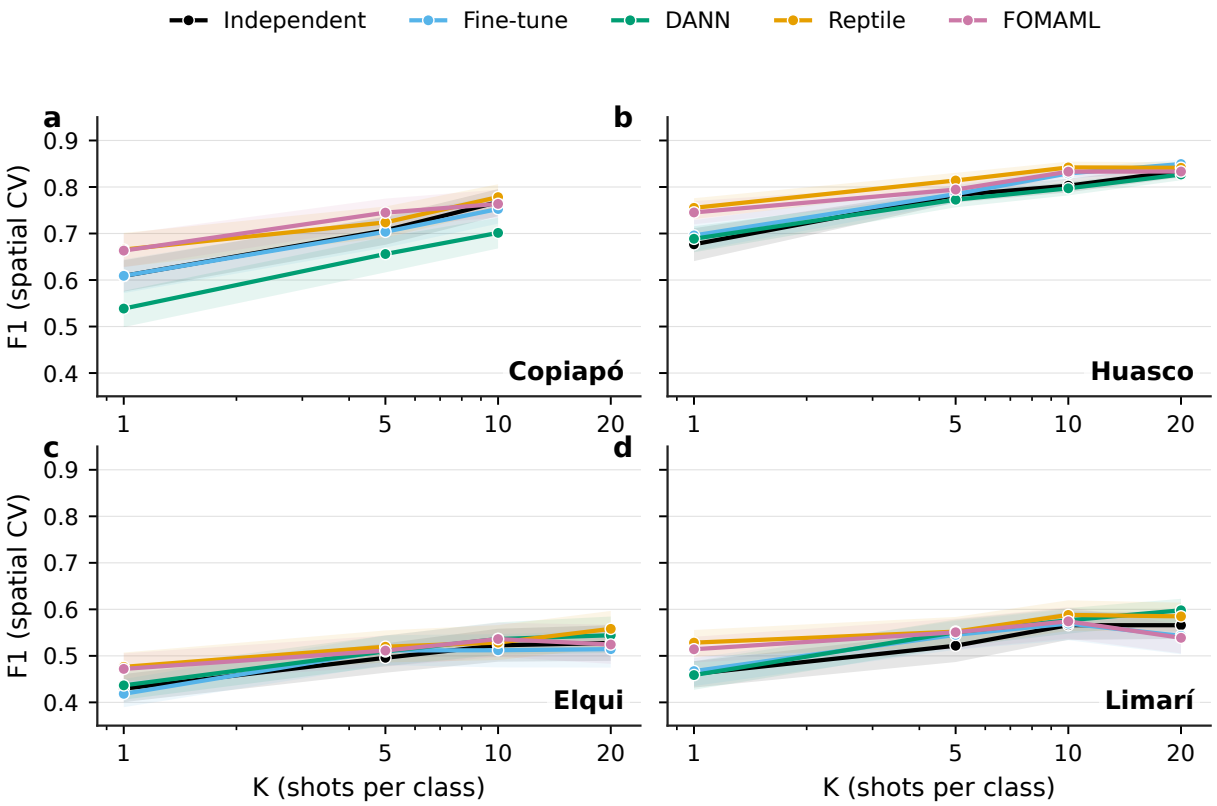


Figure A.10: F_1 versus K under spatial cross-validation, including DANN. Meta-learning (Reptile or FOMAML) beats Independent at $K = 1$ in all four target basins.

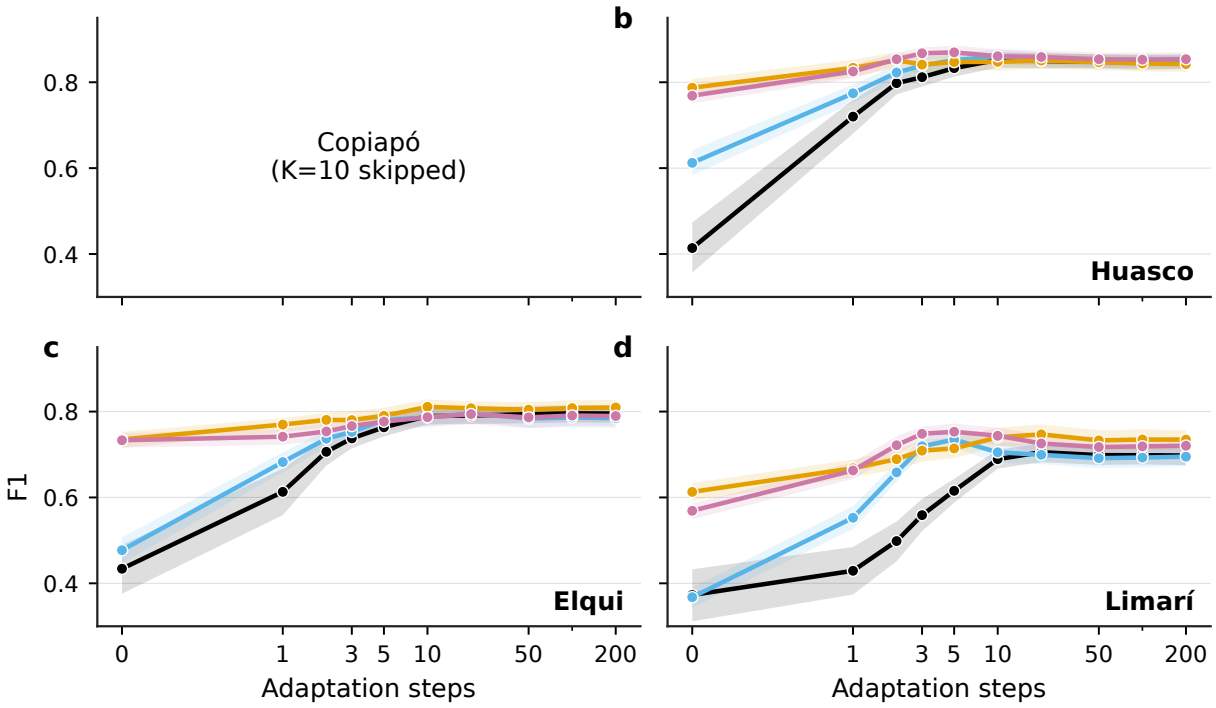


Figure A.11: Adaptation curves at $K = 10$. Same convergence pattern as $K = 5$ (Figure 4).

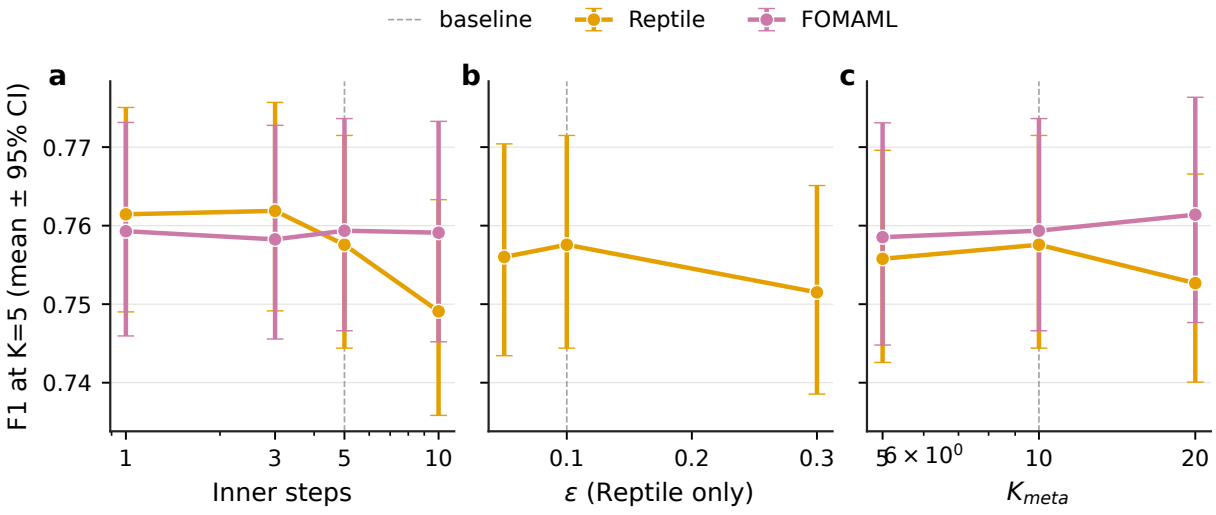


Figure A.12: Hyperparameter sensitivity ablation. (a) Inner gradient steps. (b) Reptile ϵ. (c) Meta-training K. Sensitivity is small (< 1 pp F1) — baseline configuration is robust.

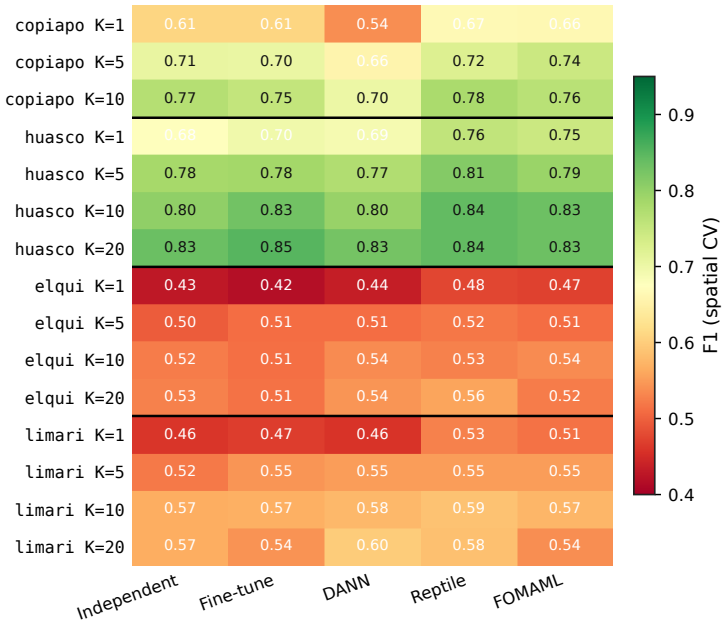


Figure A.13: F_1 (mean) for all (target $\times$ K $\times$ method) combinations under spatial CV.

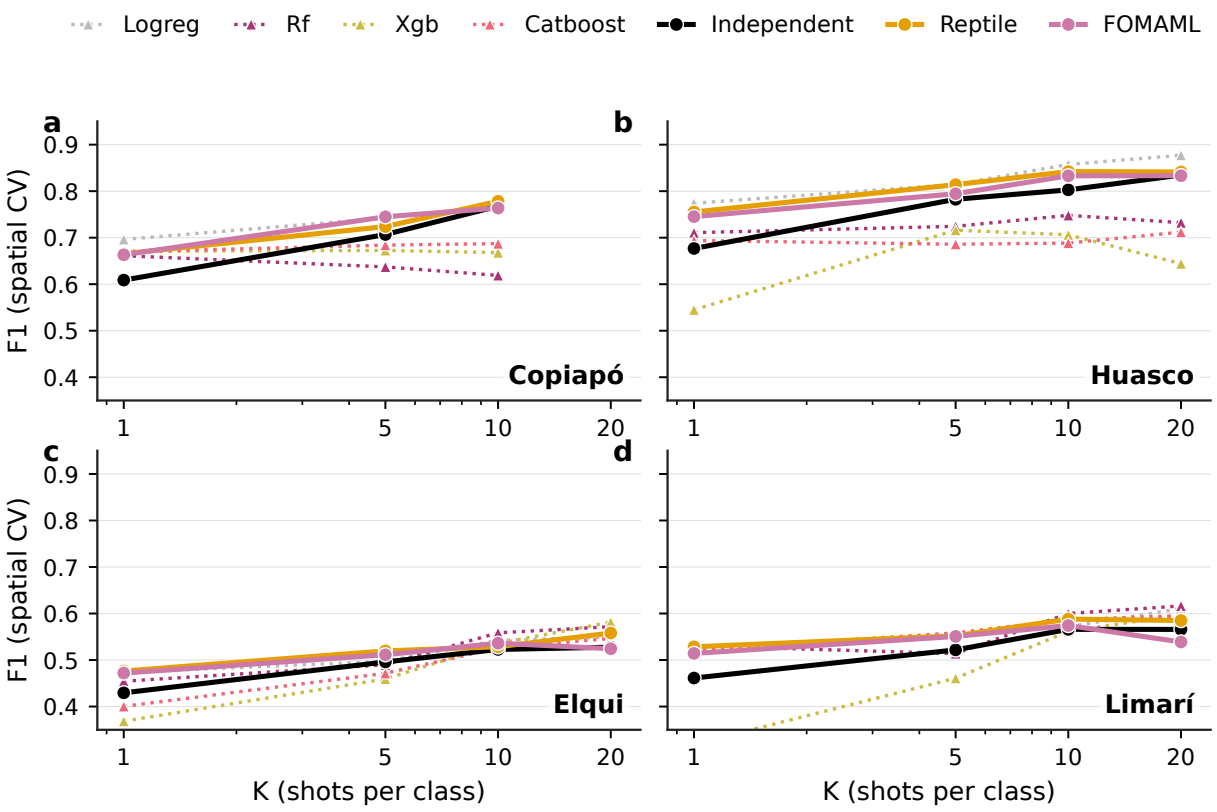


Figure A.14: Classical machine learning vs. meta-learning under spatial CV. Logistic regression, random forest, XGBoost and CatBoost are competitive with meta-learning in mean F_1, but classical models cannot adapt with $K = 0$.

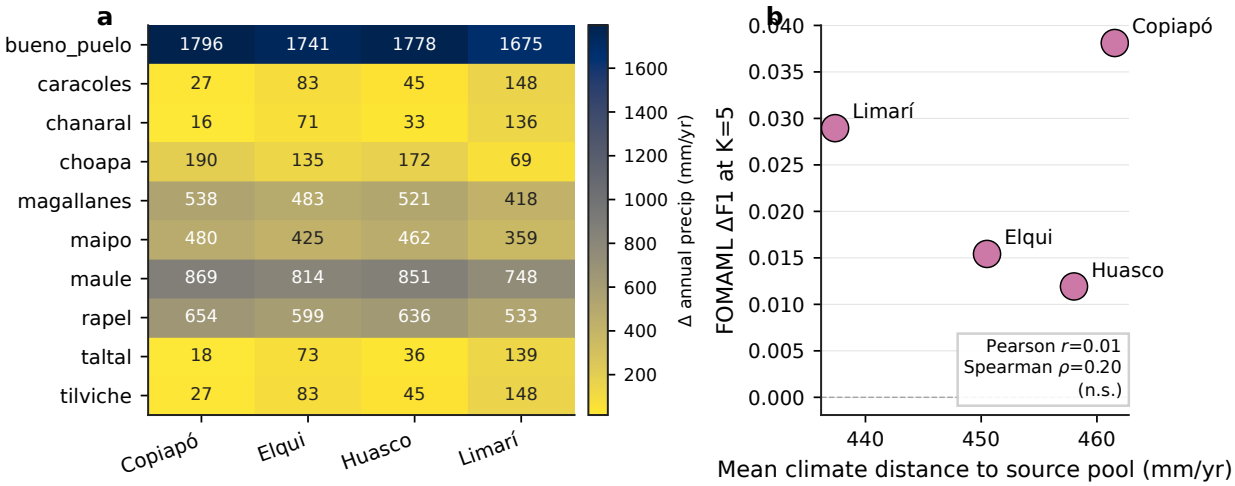


Figure A.15: (a) Climate distance matrix between source and target basins (annual precipitation difference). (b) FOMAML transfer advantage at $K = 5$ vs. mean source-target climate distance; no significant correlation (Pearson $r = 0.01$, Spearman $\rho = 0.20$).

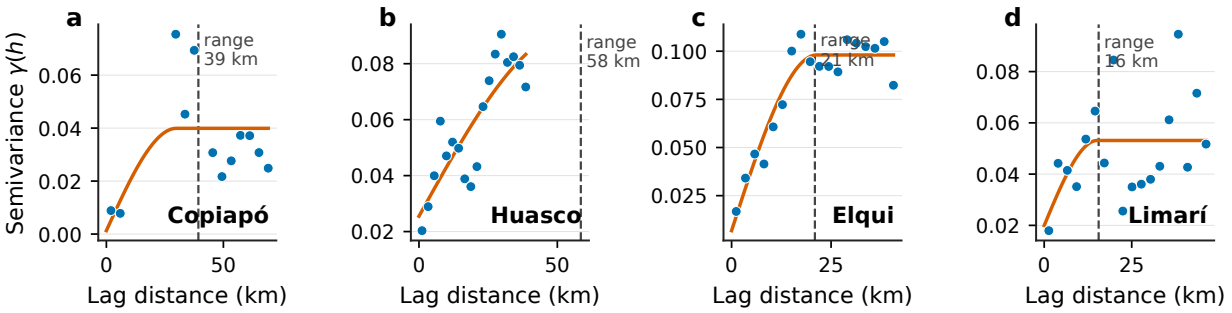


Figure A.16: Empirical variogram of Independent-model residuals for each target basin, with fitted spherical model (red dashed line marks the autocorrelation range). Fitted ranges: 39.3 km (Copiapó), 58.5 km (Huasco), 20.9 km (Elqui), 15.6 km (Limarí). At $K_{\text{folds}} = 5$ the median fold-centroid separation exceeds the range in three of four basins, confirming the spatial folds achieve genuine spatial decorrelation.

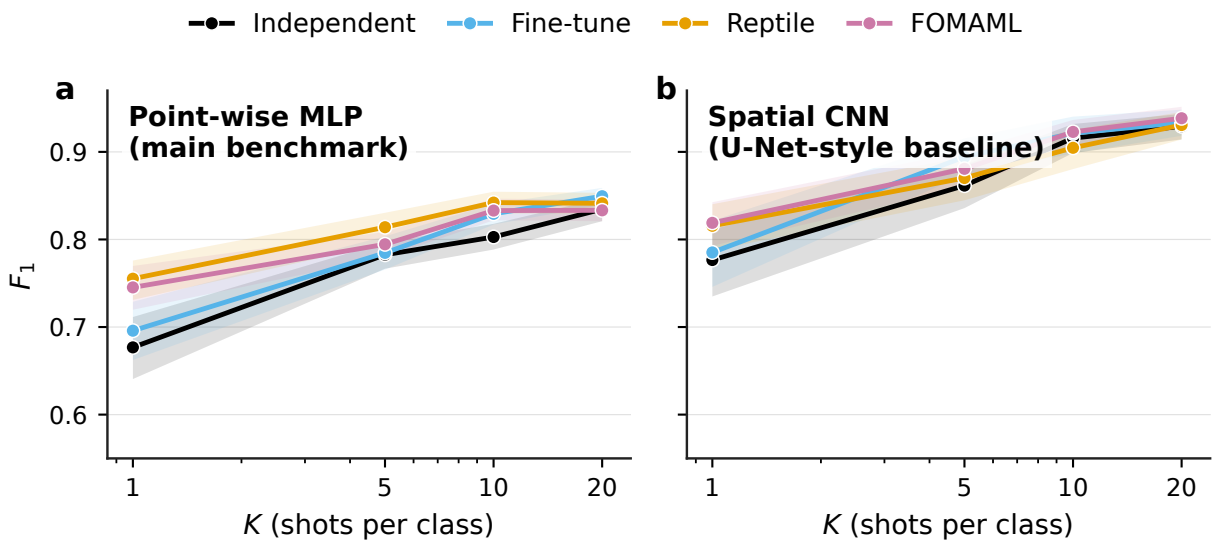


Figure A.17: Spatial CNN (U-Net-style) backbone vs. point-wise MLP on Huasco under the same spatial K-shot CV protocol with 9 source basins for meta-training. The CNN exceeds the MLP by 6–11 percentage points F_1 across all four methods and K values. The meta-learning lift over Independent is smaller in the CNN regime, indicating that spatial context absorbs part of the prediction signal that meta-learning's prior captures in the point-wise setting.

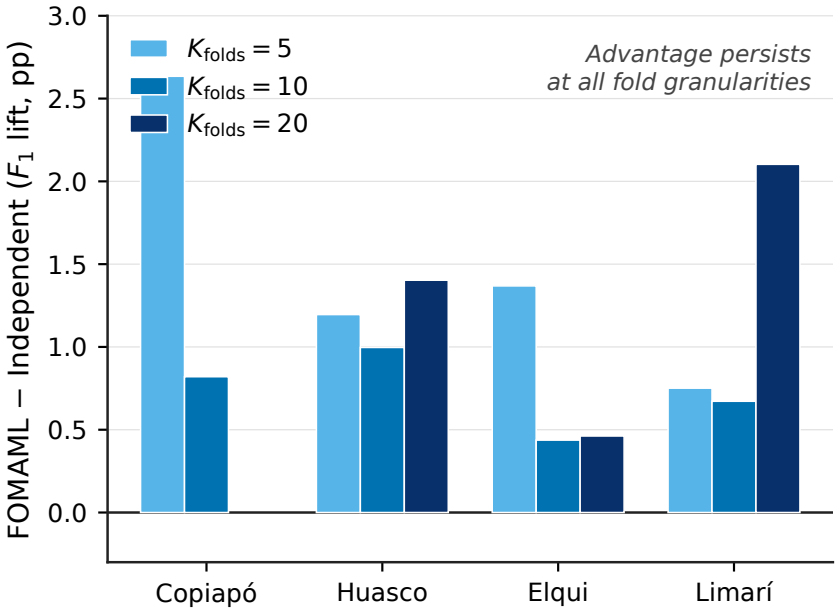


Figure A.18: FOMAML − Independent mean F_1 lift as a function of the number of spatial CV folds $K_{\text{folds}} \in \{5, 10, 20\}$. The meta-learning advantage persists at all fold granularities (per-target mean lift +0.4 to +2.6 pp, averaged over $K \in \{5, 10\}$ shots and three seeds—$K = 1$, where the advantage is largest, is not part of this experiment), demonstrating that the reported advantage is not an artifact of the $K_{\text{folds}} = 5$ choice.